

Algebraic Geometry Notes

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Introduction

Rings are always commutative with unity.

Chapter 1

Affine Space

1.1 Affine Varieties and the Nullstellensatz

In algebraic geometry we are interested in studying spaces which are locally described by polynomial equations.

Definition 1.1

Let $F \in \mathbb{R}[x, y]$. Then the **vanishing set** or zero locus of F is

$$V(F) = \{(x, y) \in \mathbb{R}^2 : F(x, y) = 0\}$$

This extends the construction of many familiar spaces, such as the circle:

$$S^1 = V(f), \quad f(x, y) = x^2 + y^2 - 1$$

These equations can give information about the spaces themselves. For instance, we say that a point in the vanishing set is *singular* if $\nabla F = 0$ at that point. For S^1 , there are no singular points, while for the more complicated space

$$y^2 - x^2(x + 1) = 0$$

there is a singular point at $(0, 0)$, which is seen to be a crossing point of this curve. Thus, we intuitively see that singular points should correspond somehow to a failure to be smooth or a manifold.

This corresponds to the general situation in algebraic geometry, where we are able to apply tools from commutative algebra to analyze our spaces, as well as giving geometric context and intuition for problems in algebra.

It is worth noting that the solution sets to polynomial equations depend on the space we work over. For instance, $x^2 + y^2 - 1 = 0$ is satisfied at four discrete points over $\mathbb{Z}^2$, a circle over $\mathbb{R}^2$, and over $\mathbb{C}^2$ the solution set is basically a sphere minus one point. One could also

work over finite fields, and so on. For now we will be working over algebraically closed fields, in particular $\mathbb{C}$.

Definition 1.2

Let K be an algebraically closed field. Then we define the **affine n -space** over K to be

$$\mathbb{A}_K^n = \{(a_1, \dots, a_n) : a_i \in K\}$$

We may also denote affine n -space by $\mathbb{A}^n$ when K is obvious.

Definition 1.3

Let $S \subseteq K[x_1, \dots, x_n]$ be a collection of polynomials. Then their **vanishing locus** is

$$V(S) = \{x \in \mathbb{A}^n : f(x) = 0 \text{ for all } f \in S\} \subseteq \mathbb{A}^n$$

Any set of this form is called an **affine variety**.

Proposition 1.1

If $S_1 \subseteq S_2 \subseteq K[x_1, \dots, x_n]$ then $V(S_1) \supseteq V(S_2)$.

Proposition 1.2

If f, g are polynomials then

$$V(f, g) = V(f) \cap V(g)$$

and

$$V(fg) = V(f) \cup V(g)$$

More generally, for collections $S_1, S_2 \subseteq K[x_1, \dots, x_n]$,

$$V(S_1 \cup S_2) = V(S_1) \cap V(S_2)$$

and

$$V(S_1 S_2) = V(S_1) \cup V(S_2)$$

where $S_1 S_2$ is the pointwise product

$$S_1 S_2 = \{fg : f \in S_1, g \in S_2\}$$

Even more generally, we can take arbitrary unions of collections $\bigcup_{\lambda \in \Lambda} S_\lambda$ (equivalently, arbitrary intersections of the varieties $\bigcap_{\lambda \in \Lambda} V(S_\lambda)$), but only *finite* products (equivalently, unions).

We note that the pointwise product result requires the fact that K is a domain (since we need the fact that $fg = 0 \implies f = 0$ or $g = 0$).

Proposition 1.3

If $X \subseteq \mathbb{A}^n, Y \subseteq \mathbb{A}^m$ are affine varieties, then $X \times Y \subseteq \mathbb{A}^{n+m}$ is also an affine variety.

Proof. Let $S_X \in K[x_1, \dots, x_n]$ and $S_Y \in K[y_1, \dots, y_m]$ be such that $X = V(S_X), Y = V(S_Y)$. Then project these naturally into $K[x_1, \dots, x_{n+m}]$ by converting each formal variable y_i to x_{n+i} , and it follows that $X \times Y = V(S'_X, S'_Y)$. $\square$

Example 1.1

In $\mathbb{A}^1$, the only affine varieties are $\mathbb{A}^1$, and finite sets. This can be seen by noting that we immediately have

$$V(S) \subseteq V(f)$$

whenever $f \in S$. Then, either S only contains the zero polynomial, or else f can be chosen to be nonzero, hence to have finitely many roots.

This logic also shows that any variety in $\mathbb{A}^1$ can be described by a single polynomial equation $(z - z_1) \cdots (z - z_n) = 0$, or the zero polynomial. (For general S this polynomial need not actually be in S , but if S is an ideal then it is.) We would like to have a similar finiteness result for varieties in $\mathbb{A}^n$. The following will demonstrate this.

Definition 1.4

Recall that if $S \subseteq K[x_1, \dots, x_n]$, then the generated **ideal** is

$$\langle S \rangle = \{r_1 f_1 + \dots + r_m f_m : r_i \in K[x_1, \dots, x_n], f_i \in S\}$$

Proposition 1.4

$$V(S) = V(\langle S \rangle).$$

Proof. To show that $V(\langle S \rangle) \subseteq V(S)$, we just note that $S \subseteq \langle S \rangle$.

To see that $V(S) \subseteq V(\langle S \rangle)$, at any element $x \in V(S)$, each $f \in S$ vanishes, so any polynomial in $\langle S \rangle$ vanishes as well. $\square$

Theorem 1.5: Hilbert's Basis Theorem

If a ring R is Noetherian then $R[x]$ is as well.

Proof. See Section A.1. $\square$

In particular note that fields are always Noetherian, so we see that $\langle S \rangle$ itself is finitely generated.

Corollary 1.6

Any affine variety can be written as $X = V(f_1, \dots, f_m) \subseteq \mathbb{A}^m$.

Thus we see that the theory of varieties can be written entirely in terms of ideals, which allows us to enjoy finiteness properties.

Definition 1.5

If J is an ideal in R then the **radical** of J is

$$\sqrt{J} = \{x \in R : \exists m \text{ s.t. } x^m \in J\}$$

Lemma 1.7

For I, J two ideals, $\sqrt{IJ} = \sqrt{I \cap J} = \sqrt{I} \cap \sqrt{J}$.

Proof. First, we have $IJ \subseteq I \cap J$, since $SI \subseteq I$ for any subset $S \subseteq R$ and any ideal I . Therefore $\sqrt{IJ} \subseteq \sqrt{I \cap J}$.

If $x \in \sqrt{I \cap J}$ with $x^m = z$, $z \in I \cap J$, then plainly $x \in \sqrt{I}, \sqrt{J}$ as well so $\sqrt{I \cap J} \subseteq \sqrt{I} \cap \sqrt{J}$.

Lastly, if $x \in \sqrt{I} \cap \sqrt{J}$, then $x^m = y \in I$ and $x^n = z \in J$ for some y, z . It follows that $x^{m+n} = yz \in IJ$. So $\sqrt{IJ} \subseteq \sqrt{I} \cap \sqrt{J}$, completing the circular inclusion. $\square$

Lemma 1.8

If J, J_1, J_2 are ideals in $K[x_1, \dots, x_n]$, then

- (a) $V(\sqrt{J}) = V(J)$.
- (b) $V(J_1) \cup V(J_2) = V(J_1 J_2) = V(J_1 \cap J_2)$.
- (c) $V(J_1) \cap V(J_2) = V(J_1 + J_2)$.

Proof. (a) $V(\sqrt{J}) \subseteq V(J)$ because $J \subseteq \sqrt{J}$. In the other direction, if $x \in V(J)$, then for any $g \in \sqrt{J}$, $g^m(x) = 0$ for some m . Since K is a field, $g(x) = 0$ so $x \in V(\sqrt{J})$.

(b) The first equality is a special case of the version for subsets. The second follows from the lemma $\sqrt{IJ} = \sqrt{I \cap J}$.

(c) This is the previous version, except that we note that $J_1 + J_2 = \langle J_1 \cup J_2 \rangle$. $\square$

At this point, we have constructed a mapping that takes ideals to affine varieties. To better understand this correspondence, we now construct the inverse map.

Definition 1.6

Let $X \subseteq \mathbb{A}^n$ be a subset. Then **vanishing ideal** of X is

$$I(X) = \{f \in K[x_1, \dots, x_n] : f|_X = 0\}$$

For similar reasons as before, this is indeed an ideal. In particular it is a radical ideal. Also, the map $X \mapsto I(X)$ is inclusion-reversing, so if $X_1 \subseteq X_2$ then $I(X_1) \supseteq I(X_2)$.

While $X \mapsto I(X)$ and $J \mapsto V(J)$ are maps from sets to ideals and vice versa, we need to clarify the manner in which they compose, since they are not strictly inverses of each other.

Theorem 1.9: Nullstellensatz

- (a) If $J \subseteq K[x_1, \dots, x_n]$ is an ideal then $I(V(J)) = \sqrt{J}$.
- (b) If $X \subseteq \mathbb{A}^n$ is an affine variety then $V(I(X)) = X$.

Proof. (a) It is clear that any f with $f^m \in J$ must vanish on $V(J)$ by our previous work, so $\sqrt{J} \subseteq I(V(J))$. The other direction is the actually difficult part of this proof and is not demonstrated. It is equivalent to the statement that if f vanishes on $V(J)$ then $f^m \in J$ for some m .

- (b) Suppose $X = V(J)$. Then $J \subseteq \sqrt{J} = I(X)$ (by part a) so $V(I(X)) \subseteq X$. In the other direction, $I(X)$ is defined to be functions which vanish on X , so necessarily X lies in their vanishing locus. $\square$

This shows that if we restrict the previous mapping between ideals and varieties to only the radical ideals, then we get a bijection. This bijection is inclusion-reversing.

Corollary 1.10

There is a bijective correspondence between maximal ideals and points $a = (a_1, \dots, a_n) \in \mathbb{A}^n$.

Proof. Maximal ideals are maximal among radical ideals (since the radical of a proper ideal is proper). By the inclusion reversing bijection, they correspond to *minimal* nonempty varieties, which are singular points. $\square$

In particular the bijection can be written as $a \mapsto \langle x_1 - a_1, \dots, x_n - a_n \rangle$.

Lemma 1.11

Let $X_1, X_2 \subseteq \mathbb{A}^n$ be affine varieties. Then

- (a) $I(X_1 \cup X_2) = I(X_1) \cap I(X_2)$,
- (b) $I(X_1 \cap X_2) = \sqrt{I(X_1) + I(X_2)}$.

Proof. (a) By the inclusion reversing bijection,

$$\begin{cases} X_1 \subseteq X_1 \cup X_2 \\ X_2 \subseteq X_1 \cup X_2 \end{cases} \implies I(X_1 \cup X_2) \subseteq I(X_1) \cap I(X_2)$$

and any function which vanishes on X_1, X_2 vanishes on their union, so $I(X_1) \cap I(X_2) \subseteq I(X_1 \cup X_2)$.

(b) We have

$$I(X_1 \cap X_2) = I(V(I(X_1)) \cap V(I(X_2))) = I(V(I(X_1) + I(X_2))) = \sqrt{I(X_1) + I(X_2)} \quad \square$$

Example 1.2

Writing $z = x + iy$ in $\mathbb{A}_{\mathbb{C}}^1$, let X_1, X_2 be the subsets $\{x = 0\}, \{y = 0\}$, respectively. Note that as before these are *not* algebraic varieties, so the above Lemma does not hold. In fact it fails, since the vanishing ideal of both is the zero ideal, but their intersection is the origin, so:

$$I(X_1 \cap X_2) = \langle x \rangle \neq 0 = \sqrt{I(X_1) + I(X_2)}$$

Corollary 1.12: Weak Nullstellensatz

If $J \subseteq K[x_1, \dots, x_n]$ is an ideal, then the following are equivalent:

- (a) $V(J) = \emptyset$,
- (b) $J = K[x_1, \dots, x_n]$,
- (c) $1 \in J$.

Proof. (b) $\iff$ (c) is basic ring theory. Now note that if $1 \in J$ then clearly $V(J) = \emptyset$, and if $V(J) = \emptyset$ then 1 vanishes on $V(J)$, so $1 \in \sqrt{V(J)} \implies 1 \in J$. $\square$

1.2 Subvarieties

We now study varieties which live inside of other varieties, instead of simply $\mathbb{A}^n$.

Definition 1.7

Let $X \subseteq \mathbb{A}^n$ be an affine variety. Then the **coordinate ring** of X is

$$A(X) = \{f|_X : f \in K[x_1, \dots, x_n]\} \subseteq K^{(K^n)}$$

Note that the elements of $A(X)$ are considered to be functions represented by polynomials, not polynomials, and only need to be equal as functions. For instance, $x^2 + x$ is the zero function on $\mathbb{F}_2$, but is not equal as a polynomial to the zero polynomial. ($\mathbb{F}_2$ is not algebraically closed, but this is just for illustration.)

Lemma 1.13

$A(X)$ is a K -algebra.

Recall that a K -algebra can equivalently be defined either as a K -vector space with a bilinear product, or as a ring A with a ring homomorphism from K into $Z(A)$.

Proof. $A(X)$ is clearly a ring, and the homomorphism takes $\lambda \in K$ to the constant polynomial $f \equiv \lambda$. $\square$

Example 1.3

If $X \subseteq \mathbb{A}^1$ is a finite set with r points, then by Lagrange interpolation we can define a polynomial with any prescribed values on these points. So $A(X) = K^r$.

Proposition 1.14

Let $X \subseteq \mathbb{A}^n$ be an affine variety. Then $f \mapsto f|_X$ is a surjective ring homomorphism with kernel $I(X)$, so that

$$A(X) = K[x_1, \dots, x_n] / I(X)$$

Proof. Surjectivity is by definition. Then, we have injectivity since $f|_X = 0$ if and only if f vanishes on X , which is to say that $f \in I(X)$. $\square$

In particular if $X = V(J)$ then

$$A(V(J)) = K[x_1, \dots, x_n] / \sqrt{J}$$

Corollary 1.15

$$A(\mathbb{A}^n) = K[x_1, \dots, x_n].$$

Proof. We have $\mathbb{A}^n = V(\langle 0 \rangle)$, so $I(\mathbb{A}^n) = I(V(\langle 0 \rangle)) = \sqrt{\langle 0 \rangle} = \langle 0 \rangle$. Hence

$$A(\mathbb{A}^n) = K[x_1, \dots, x_n]$$

$\square$

This shows that the theory of polynomials on $\mathbb{A}^n$ is really the theory of polynomial *functions* on $\mathbb{A}^n$, so there is no generality of studying polynomial functions on varieties. In this way, we can see that our previous geometric correspondence from ideals in $A(\mathbb{A}^n)$ should also give some meaning to ideals in general $A(X)$. These turn out to be the subvarieties.

Definition 1.8

Fix an affine variety $Y \subseteq \mathbb{A}^n$. Then for any $S \subseteq A(Y)$, define its **vanishing locus** to be

$$V(S) = V_Y(S) = \{x \in Y : f(x) = 0 \forall f \in S\}$$

A subset which is the vanishing locus of some $S \subseteq A(Y)$ is called an **affine subvariety**.

Correspondingly, for any subset $X \subseteq Y$, we define the **vanishing ideal** of X in Y to be

$$I(X) = I_Y(X) = \{f \in A(Y) : f|_X \equiv 0\}$$

This new correspondence behaves almost exactly the same as the one for varieties in $\mathbb{A}^n$.

Proposition 1.16

Let $Y \subseteq \mathbb{A}^n$ be an affine variety.

- (a) $I_Y(X)$ is a (radical) ideal,
- (b) any subvariety $X \subseteq Y$ is a variety in $\mathbb{A}^n$,
- (c) any variety $X \subseteq \mathbb{A}^n$ such that $X \subseteq Y$ is also an affine subvariety of Y ,
- (d) whenever $X \subseteq Y$ is a subvariety,

$$A(X) \cong A(Y)/I(X)$$

- (e) I_Y, V_Y give an inclusion reversing, bijective correspondence between affine subvarieties of Y and radical ideals of $A(Y)$,
- (f) $V(J_1) \cup V(J_2) = V(J_1 J_2) = V(J_1 \cap J_2)$.
- (g) $V(J_1) \cap V(J_2) = V(J_1 + J_2)$.
- (h) $I(X_1 \cup X_2) = I(X_1) \cap I(X_2)$,
- (i) $I(X_1 \cap X_2) = \sqrt{I(X_1) + I(X_2)}$.

Proof. (a) Clear by lifting to polynomials.

- (b) Let $X = V_Y(J)$ for an ideal $J \subseteq A(Y)$. Let π be the quotient map from $K[x_1, \dots, x_n]$ by $I(Y)$, and let $I = \pi^{-1}(J)$. Then since $X \subseteq Y$, for $x \in X$, $\pi g(x) = 0$ if and only if $g(x) = 0$. So

$$\begin{aligned} X = V_Y(J) &= \{x : g'(x) = 0 \forall g' \in J\} = \{x : \pi g(x) = 0 \forall g \in \pi^{-1}(J)\} \\ &= \{x : g(x) = 0 \forall g \in \pi^{-1}(J)\} = V(\pi^{-1}(J)) \end{aligned}$$

and X is a variety.

- (c) Let $X = V(J)$. Then by the same logic as above,

$$\begin{aligned} X = V(J) &= \{x : g(x) = 0 \forall g \in J\} = \{x : \pi g(x) = 0 \forall g \in J\} \\ &= \{x : g'(x) = 0 \forall g' \in \pi(J)\} = V_Y(\pi(J)) \end{aligned}$$

so X is an affine subvariety of Y .

(d)

□

If we are given two varieties $X \subseteq \mathbb{A}^n, Y \subseteq \mathbb{A}^m$ which are cut out by

$$I(X) \subseteq K[x_1, \dots, x_n], \quad I(Y) \subseteq K[y_1, \dots, y_m]$$

then we know the product is a variety $\mathbb{A}^{n+m}$, which is cut out by taking

$$I_{X \times Y} = I(X)K[x_1, \dots, x_n, y_1, \dots, y_m] + I(Y)K[x_1, \dots, x_n, y_1, \dots, y_m]$$

This is a reflection of a more general fact concerning tensor products, since

$$K[x_1, \dots, x_n, y_1, \dots, y_m] = K[x_1, \dots, x_n] \otimes_K K[y_1, \dots, y_m]$$

Proposition 1.17

Let $I_A \subseteq A, I_B \subseteq B$ be ideals, and let C be a ring with morphisms into A, B . Then

$$I_{A,B} = I_A(A \otimes_C B) + I_B(A \otimes_C B)$$

is an ideal of $A \otimes_C B$, and the map

$$\begin{aligned} (A/I_A) \otimes_C (B/I_B) &\rightarrow A \otimes_C B / I_{A,B} \\ (a + I_A) \otimes_C (b + I_B) &\mapsto a \otimes_C b + I_{A,B} \end{aligned}$$

is an isomorphism.

Proposition 1.18

Let $X \subseteq \mathbb{A}^n, Y \subseteq \mathbb{A}^m$ with $I_{X \times Y}$ defined as above. Then $I(X \times Y) = I_{X \times Y}$ and

$$A(X \times Y) = A(X) \otimes_K A(Y)$$

Proof. It is clear that $V(I_{X \times Y}) = X \times Y$, so by the Nullstellensatz,

$$I(X \times Y) = \sqrt{I_{X \times Y}}$$

So we need to show $I_{X \times Y}$ is radical, which is equivalent to saying that

$$K[x_1, \dots, x_n, y_1, \dots, y_m] / I_{X \times Y}$$

is reduced. By the previous proposition we have that

$$\begin{aligned} &K[x_1, \dots, x_n, y_1, \dots, y_m] / I_{X \times Y} \\ &\cong \left(K[x_1, \dots, x_n] / I(X) \right) \otimes_K \left(K[y_1, \dots, y_m] / I(Y) \right) = A(X) \otimes_K A(Y) \end{aligned}$$

This proves the second claim contingent on the first, but also we know that $A(X), A(Y)$ are reduced since $I(X), I(Y)$ are radical. Then the fact that $A(X) \otimes_K A(Y)$ is reduced follows from the general fact that $U \otimes_K V$ is reduced whenever U, V are reduced K -algebras for K a perfect field. $\square$

1.3 The Zariski Topology

We previously saw that arbitrary intersections and finite unions of affine varieties were again varieties. This is the same property as the closed subsets of a topological space, which leads us to define the Zariski topology on $\mathbb{A}^n$. One immediate goal with this definition will be to decompose varieties into their irreducible components.

Definition 1.9

The **Zariski topology** on $\mathbb{A}^n$ is the topology whose closed sets are the affine varieties in $\mathbb{A}^n$. The Zariski topology on a variety $X \subseteq \mathbb{A}^n$ is the subspace topology of $\mathbb{A}^n$, whose closed sets are the subvarieties of X .

Example 1.4

The Zariski topology on $\mathbb{A}^1$ is the cofinite topology.

The Zariski topology is fairly pathological as a topological space.

Example 1.5

Let $a_1, a_2, \dots$ be a non-repeating sequence of points in $\mathbb{A}^1$. Fix $a \in \mathbb{A}^1$ and an open neighborhood U containing a . Then the complement of U is finite, so eventually every point in the sequence lies in U . Therefore (a_i) converges to every point of $\mathbb{A}^1$.

Example 1.6

Let $f : \mathbb{A}^1 \rightarrow \mathbb{A}^1$ be injective. Then any closed set is finite, and thus has a finite preimage, which is closed. So f is continuous.

Example 1.7

The Zariski topology on $\mathbb{A}^2$ is not the product topology on $\mathbb{A}^1 \times \mathbb{A}^1$. To see this, let

$$\Delta = \{(a, a) : a \in K\} \subseteq \mathbb{A}^2$$

Δ is clearly closed in the Zariski topology on $\mathbb{A}^2$. If Δ is closed in the product topology, then the complement can be written as the union of rectangular sets:

$$\mathbb{A}^2 \setminus \Delta = \bigcup_i (U_i \times V_i), \quad U_i, V_i \in \mathcal{O}(\mathbb{A}^1)$$

But for any U, V open, U, V are cofinite and $\mathbb{A}^1$ is infinite, so there is $x \in U \cap V$. This means that $(x, x) \in U \times V$, contradicting the decomposition of $\mathbb{A}^2 \setminus \Delta$.

Proposition 1.19

A space X is Hausdorff if and only if the diagonal is closed in the product topology on $X \times X$.

Proof. We want to show that $\Delta^c = (X \times X) \setminus \Delta$ is open. Let $(x, y) \in \Delta^c$. Since X is Hausdorff there are $U \ni x, V \ni y$ disjoint, so that $U \times V$ contains (x, y) and does not intersect the diagonal. Unioning over all pairs in Δ^c , Δ^c is open so Δ is closed. The same logic works in the other direction. $\square$

This gives a faster proof that the Zariski topology is not the product topology. Also, we will later see that this extends to higher products $\mathbb{A}^{n+m}$, since $U \cap V$ will always be nonempty for appropriate U, V , even outside the cofinite topology.

Definition 1.10

Let X be a topological space. X is **reducible** if it can be written as the union $X = X_1 \cup X_2$ for two proper closed subsets. It is **disconnected** if it can be written as the union of two nonempty disjoint closed subsets.

If X is disconnected then it is reducible, so if X is irreducible then it is connected.

Proposition 1.20

Let X be irreducible. Then
 (a) Any two nonempty open sets have nonempty intersection.
 (b) Any nonempty open set is dense in X .

Proof. (a) If U_1, U_2 are nonempty and open with empty intersection, then $X \setminus U_1, X \setminus U_2$ are proper closed subsets whose union is all of X .

(b) A set is dense if and only if it intersects every nonempty open set, so this follows from (a). $\square$

We want ways to describe the irreducible and connected ideals cleanly. Topologically this is difficult, so we will convert their definitions to algebraic ones.

Proposition 1.21

A nonempty affine variety $X \subseteq \mathbb{A}^n$ is irreducible if and only if $A(X)$ is a domain (which is equivalent to $I(X)$ being a prime ideal).

Proof. ($\implies$) Let X be irreducible, take $f, g \in A(X)$ with $fg = 0$, and let $X_1 = V_X(\langle f \rangle)$, $X_2 = V_X(\langle g \rangle)$. Then $X = X_1 \cup X_2$ which are both closed, so $X_1 = X$ or $X_2 = X$. In the first case $f = 0$ and in the second $g = 0$.

($\impliedby$) Let $X = X_1 \cup X_2$ with X_1, X_2 closed. Then X_1, X_2 are both subvarieties of X , so let $I_1 = I_X(X_1)$, $I_2 = I_X(X_2)$. Then $I_1 I_2$ is the zero ideal in $A(X)$, which implies that at least one is zero. But if I_1 is zero then $X_1 = X$ and similarly for I_2 . $\square$

This tells us that the Nullstellensatz restricts to a bijective correspondence between nonempty irreducible affine subvarieties of $Y \subseteq \mathbb{A}^n$ to prime ideals of $A(Y)$.

Example 1.8

A finite set is irreducible if and only if it is a single point. In this case $A(X) = K$ which is a domain.

Proposition 1.22

$\mathbb{A}^n$ is irreducible (hence connected).

Proof. $A(\mathbb{A}^n) = K[x_1, \dots, x_n]$ is an integral domain. $\square$

1.4 Irreducible Decompositions

To show that we can decompose any affine variety into irreducible components, we will prove this for more general topological spaces. This will allow us to easily extend to other forms of varieties later.

Definition 1.11

A topological space X is **Noetherian** if there is no infinite strictly decreasing chain of closed subsets

$$X_0 \supsetneq X_1 \supsetneq X_2 \supsetneq \dots$$

Recall that Noetherian rings satisfy the *ascending* chain condition on ideals; here we have a *descending* chain condition since we work over closed (equivalently, a Noetherian space satisfies the ascending condition on opens).

Proposition 1.23

Quotients of Noetherian rings are Noetherian.

Proof. Ideals in the quotient correspond to ideals in the original ring (containing the quotiented ideal), so an ascending chain in the quotient lifts to an ascending chain in the original. $\square$

Lemma 1.24

An affine variety $X \subseteq \mathbb{A}^n$ is Noetherian.

Proof. Suppose there is a descending chain

$$X_0 \supsetneq X_1 \supsetneq X_2 \supsetneq \dots$$

Then by the inclusion reversing bijection, we get

$$I(X_0) \subsetneq I(X_1) \subsetneq \dots \subseteq A(X)$$

But $A(X)$ is a quotient of $K[x_1, \dots, x_n]$, which is Noetherian, hence $A(X)$ is Noetherian. This is a contradiction. $\square$

Proposition 1.25

Any subset of a Noetherian space is Noetherian under the subspace topology.

Proof. Let $X \subseteq Y$ with Y Noetherian. Suppose there is some descending chain

$$X_0 \supsetneq X_1 \supsetneq X_2 \supsetneq \dots$$

of relatively closed sets in X . Then by definition there are $Y_i \subseteq Y$ closed such that $Y_i \cap X = X_i$. It follows that

$$Y_0 \supseteq (Y_0 \cap Y_1) \supseteq \dots$$

is a descending chain (not necessarily strict) in Y . To see that this is closed, we have

$$\left(\bigcap_{i=0}^n Y_i \right) \cap X = \bigcap_{i=0}^n (Y_i \cap X) = \bigcap_{i=0}^n X_i = X_n$$

so we actually have

$$Y_0 \supsetneq (Y_0 \cap Y_1) \supsetneq \dots$$

This is a strict descending chain of closed sets in Y , contradicting the assumption that it is Noetherian. $\square$

Proposition 1.26

Any Noetherian topological space X can be written as the finite union of nonempty irreducible closed subsets

$$X = \bigcup_{i=1}^r X_i$$

If we also assume that $X_i \not\subseteq X_j$ for $i \neq j$, then this is unique up to permutation.

Proof. If X is empty then we take $r = 0$ and are done. Otherwise, assume that X is not a finite union of irreducibles. Then X itself is certainly not irreducible, so there are proper nonempty closed subsets X_1, X_2 with $X_1 \cup X_2 = X$. These cannot both be irreducible, so one (say X_1) can also be written as the union of two proper nonempty closed subsets. This gives a strictly descending chain of closed subsets, contradicting Noetherianity. So X must have been a finite union of irreducibles.

To see uniqueness, take two decompositions

$$X = \bigcup_{i=1}^r X_i = \bigcup_{j=1}^r Y_j$$

Then for any X_i we can intersect against the other decomposition and apply irreducibility:

$$X_i = \bigcup_{j=1}^r (Y_j \cap X_i)$$

By irreducibility there is j such that $Y_j \cap X_i = X_i$, or $X_i \subseteq Y_j$. Then applying the decomposition in the other direction we get i' s.t. $Y_j \subseteq X_{i'}$. But $X_i \subseteq X_{i'}$ is only satisfied by $i = i'$. Thus we have $X_i = Y_j$, giving a bijection between the decompositions, so they are unique up to permutation. $\square$

Now that we have proven that X can be decomposed into irreducible components in the topological sense, we return to affine varieties and the question of how to actually compute this decomposition.

Definition 1.12

A **primary ideal** I of a ring R is an ideal such that whenever $xy \in I$, either x or a power of y is in I .

Equivalently, $xy \in I$ if and only if either x or y is in I , or if both x, y are in $\sqrt{I}$.

Proposition 1.27

Radicals of primary ideals are prime.

Proof. Let $xy \in \sqrt{I}$ for I primary. Then $(xy)^m \in I \implies x^m y^m \in I$. We either have $x^m \in I$ or $y^m \in I$, or else both x^m, y^m are in $\sqrt{I}$. In every case at least one of x, y is in $\sqrt{I}$. $\square$

In Section A.2, we show that in a Noetherian ring, every ideal (in particular, $I(X)$) has a *primary decomposition*

$$I(X) = \bigcap_{i=1}^r Q_i$$

of primary ideals Q_i . Then this is

$$I(X) = \sqrt{I(X)} = \sqrt{\bigcap_{i=1}^r Q_i} = \bigcap_{i=1}^r \sqrt{Q_i}$$

since if y lies in the radical of the intersection, then y^{m_i} lies in each Q_i for some m_i , hence $y \in \sqrt{Q_i}$ for each i .

Define $P_i = \sqrt{Q_i}$. These ideals are prime since the radicals of prime ideals are prime. So

$$I(X) = \bigcap_{i=1}^r P_i \implies X = V(I(X)) = \bigcup_{i=1}^r V(P_i)$$

Each $V(P_i)$ is irreducible since P_i is prime. Discard any P_i which are contained in another P_j (i.e. those maximal w.r.t. inclusion within the decomposition). Then this is a decomposition into irreducible closed subsets, which do not contain each other, and is therefore the unique decomposition.

Proposition 1.28

If X is an affine variety, then the irreducible components X_i of X are the maximal (w.r.t. inclusion) irreducible subvarieties of X .

Proof. Suppose not. Then writing

$$X = \bigcup_{i=1}^r X_i$$

there must be some X_j which is not maximal. Suppose without loss of generality that this is X_1 . Then let $X_1 \subsetneq X'_1$ where X'_1 is also an irreducible subvariety. Then

$$X = X'_1 \cup \bigcup_{i=2}^r X_i$$

is another decomposition into irreducibles, and after possibly discarding any subvarieties contained in X'_1 , we get a distinct decomposition, contradiction. $\square$

Corollary 1.29

There is a bijection between irreducible components of X and minimal prime ideals of $A(X)$.

Note that starting from the irreducible decomposition of a Noetherian space, we can obtain a decomposition into its connected components. We do this by making a graph where each vertex is an irreducible component, and adding edges between components which intersect. Then the path connected components of the graph are the connected components of X .

As before, we would like to translate this to an algebraic condition.

Theorem 1.30: Chinese Remainder Theorem

If I_1, I_2 are ideals in R that are coprime ($I_1 + I_2 = R$), then the natural ring homomorphism

$$R/I_1 \cap I_2 \rightarrow (R/I_1) \times (R/I_2)$$

is an isomorphism.

Proposition 1.31

Suppose X is a disconnected affine variety with disjoint closed subsets $X_1 \cup X_2 = X$. Then the map $A(X) \rightarrow A(X_1) \times A(X_2)$

$$f \mapsto f|_{X_1} \times f|_{X_2}$$

is an isomorphism.

Proof. We know that

$$A(X_i) = A(X)/I_X(X_i)$$

The map given is the natural ring homomorphism

$$A(X) \rightarrow (A(X)/I(X_1)) \times (A(X)/I(X_2))$$

Since $X = X_1 \cup X_2$, we have $0 = I(X) = I(X_1) \cap I(X_2)$. Since $\emptyset = X_1 \cap X_2$ we have $A(X) = I(\emptyset) = \sqrt{I(X_1) + I(X_2)}$ so $I(X_1) + I(X_2) = A(X)$. Then we apply the Chinese remainder theorem to see that

$$A(X)/I(X_1) \cap I(X_2) = A(X) \rightarrow (A(X)/I(X_1)) \times (A(X)/I(X_2))$$

is an isomorphism. □

So we see that functions on disconnected varieties are just arbitrary combinations of functions on their connected components, so we can essentially just study the rings of functions on connected components.

1.5 Dimensions of Algebraic Varieties

So far we have not formally defined dimension for affine varieties, although examples like curves and surfaces may give us some intuition. Here, as with our previous work, we define this notion topologically and then calculate it using algebra.

Definition 1.13

Let X be a nonempty topological space. Then the **dimension** of X is the supremum over all n such that there is a chain of length n of irreducible closed subsets:

$$\emptyset \subsetneq Y_0 \subsetneq Y_1 \subsetneq \dots \subsetneq Y_n \subseteq X$$

If $Y \subseteq X$ is a closed irreducible subset then its **codimension** is the supremum over n such that there is a chain of length n of irreducible closed subsets containing Y :

$$Y \subseteq Y_0 \subsetneq Y_1 \subsetneq \dots \subsetneq Y_n \subseteq X$$

Example 1.9

The only irreducible closed subsets of $\mathbb{A}^1$ are points and $\mathbb{A}^1$, so it has dimension 1.

Note that a general Noetherian topological space is not necessarily of finite dimension:

Example 1.10

Consider $\mathbb{N}$ where the closed sets are of the form $\{1, \dots, n\}$. This is Noetherian but it has an infinite chain of irreducible closed subsets.

However, for affine varieties, the ring properties will force us to have finite dimension:

Definition 1.14

Let R be a ring. Its **Krull dimension** is the supremum over all n such that there is a chain of length n of prime ideals

$$R \supsetneq P_0 \supsetneq P_1 \supsetneq \dots \supsetneq P_n$$

The **height** of a prime ideal P is the supremum over all n such that there is a chain of length n of prime ideals containing P :

$$R \supsetneq P_0 \supsetneq P_1 \supsetneq \dots \supsetneq P_{n-1} \supsetneq P_n$$

Lemma 1.32

If X is a nonempty irreducible subvariety of an affine variety X , then

- (a) $\dim X$ is the Krull dimension of $A(X)$,

- (b) $\text{codim}_X Y$ is the height of $I_X(Y)$ in $A(X)$.
- (c) Both are finite numbers.

Proof. Any chain in $A(X)$ bijects with a chain in X . The lengths are bounded because $A(X)$ is finitely generated as a K -algebra. $\square$

Lemma 1.33

If R, S are finitely generated K -algebras which are also domains, then

$$\dim(R \otimes_K S) = \dim R + \dim S$$

We just give a sketch; the point is that we can produce a **Noether normalization** which is an injection from

$$K[x_1, \dots, x_n] \rightarrow R$$

which makes R a finitely generated module over the image (where $n = \dim R$). We do this for both R, S , giving a map

$$K[x_1, \dots, x_n, y_1, \dots, y_m] \rightarrow R \otimes_K S$$

which is a Noether normalization.

Proposition 1.34

Let X, Y be nonempty irreducible affine varieties. Then:

- (a) $\dim(X \times Y) = \dim X + \dim Y$,
- (b) If $Y \subseteq X$ then $\dim X = \dim Y + \text{codim}_X Y$,
- (c) If $f \in A(X)$ is nonzero, then every irreducible component of $V_X(f)$ has codimension 1.

Proof. (a) We know that $A(X \times Y) = A(X) \otimes_K A(Y)$. Then the result follows by the Lemma.

- (b) $A(X)$ is nice enough that all of the maximal chains of prime ideals have the same length. So then since Y is irreducible, $I_X(Y)$ is prime and the chains in X with $I_X(Y)$ as a step can be decomposed into a length $\text{codim}_X Y$ segment and a length $\dim Y$ segment.
- (c) Krull's principal ideal theorem says that every minimal prime ideal containing a (nonunit, non-zero divisor) f has height 1. $\square$

In particular, if X is irreducible then $\text{codim}_X \{a\} = \dim X$ for any point $a \in X$. Now, we develop a similar relation for reducible varieties.

Proposition 1.35

Let X be a topological space and $A \subseteq X$. Then $\dim A \leq \dim X$.

Proof. Any chain of closed irreducibles in A can be written as

$$\emptyset \subsetneq (A \cap X_0) \subsetneq \dots \subsetneq (A \cap X_n) \subseteq A$$

where the X_i are closed. The X_i can also be taken to be irreducible, and then we have that

$$\emptyset \subsetneq \overline{A \cap X_0} \subsetneq \dots \subsetneq \overline{A \cap X_n} \subseteq A$$

The inclusions are strict because we can take points in $X_n \setminus X_{n-1}$. □

Proposition 1.36

Let X be a Noetherian space and Y an irreducible subset of X . Then one of the irreducible components of X contains Y .

Proof. Suppose the decomposition is

$$X = X_1 \cup \dots \cup X_n$$

Then

$$Y = \overline{Y \cap X_1} \cup \dots \cup \overline{Y \cap X_n}$$

Since Y is irreducible, there is some i such that $\overline{Y \cap X_i} = Y$. But $\overline{Y \cap X_i} = Y \cap \overline{X_i} = Y \cap X_i$ so we have $Y \subseteq X_i$. □

Proposition 1.37

(a) If X has irreducible decomposition $X_1 \cup \dots \cup X_n$, and X is a Noetherian topological space, then

$$\dim X = \max \{\dim X_i\}$$

(b) For any topological space,

$$\dim X = \sup_{a \in X} \{\operatorname{codim}_X \{a\}\}$$

Proof. (a) We automatically have $\dim X \geq \max \{\dim X_i\}$. Now, take a chain of length $r = \dim X$:

$$\emptyset \subsetneq Y_0 \subsetneq \dots \subsetneq Y_r \subseteq X$$

Then Y_r is contained in some X_i , so we get $r \geq \dim X_i$.

(b) This is clear from the formula. □

Definition 1.15

A Noetherian topological space is of **pure dimension** if every irred. component has the same dimension. An affine variety of dimension 1 is a **curve**, of dimension 2 a **surface**, and of codimension 1 a **hypersurface**.

Chapter 2

Sheaves and Morphisms

2.1 Regular Functions on Affine Varieties

As when developing a theory of categories like topological spaces, differentiable manifolds, and so on, we would like to describe what the morphisms in the category of affine varieties should be. We will begin with functions $f : X \rightarrow \mathbb{A}^1$, since these are the mostly closely related to polynomial functions, hence easiest to describe in language related to what we have already seen. Taking continuous and differentiable maps as our examples, we note that the property of being a continuous or differentiable map is something that can be checked locally, and is preserved under restrictions to smaller domains. Moreover, if two such maps agree on their open intersection, then they “glue” into a map on the union of their domains. We will define maps from $X \rightarrow \mathbb{A}^1$ in such a way that these properties also hold.

Perhaps the most immediately obvious maps to consider are the polynomial functions. However, if we work only with maps defined on open sets, then we have access to a richer class of maps. Recall that by definition, open sets are the complement of the vanishing locus of some functions. As a result, if U is open in X and $f \in I(X \setminus U)$, then for $g \in A(X)$, the expression

$$\varphi = \frac{g}{f}$$

has nonzero denominator in U , so at least this expression makes sense to write down. In order to mimic the local and gluing properties of continuous and differentiable maps, though, we will only require that the functions are *locally* given in the form of g/f , as opposed to globally. This isn’t necessarily justification for why these are the right maps to look at, but it turns out that they are.

Definition 2.1

Let X be an affine variety and $U \subseteq X$ open. Then a **regular function** on U is a function $\varphi : U \rightarrow \mathbb{A}^1$ such that for every $a \in U$ there is $f, g \in A(X)$ and an open

subset $a \in U_a \subseteq U$ such that

$$\varphi(x) = \frac{g(x)}{f(x)}$$

on U , with $f(x) \neq 0$ on U . We denote the set of regular functions on U by $\mathcal{O}_X(U)$. This carries the structure of a K -algebra.

Example 2.1

Let $X = V(x_1x_4 - x_2x_3)$ and let $U = X \setminus V(x_2, x_4)$. Then define

$$\varphi(x_1, x_2, x_3, x_4) = \begin{cases} \frac{x_1}{x_2}, & x_2 \neq 0 \\ \frac{x_3}{x_4}, & x_4 \neq 0 \end{cases}$$

When x_2, x_4 both nonzero then $x_1x_4 - x_2x_3$ implies that this is well defined. But neither quotient makes sense on all of $V(x_1x_4 - x_2x_3)$ (and more generally, there is no global representation).

Lemma 2.1

Let $U \subseteq X$ be open and let $\varphi \in \mathcal{O}_X(U)$. Then

$$V(\varphi) = \{x \in U : \varphi(x) = 0\}$$

is closed in U .

Proof. We recall that if $\{U_i : i \in I\}$ is an open cover of U , then $V \subseteq U$ is closed in U if and only if $V \cap U_i$ is closed in U_i for each i . Now, for each $a \in U$ we have $U_a \subseteq U$ open and f, g with $\varphi = g/f$ on U_a . Since f is nonvanishing, we have

$$V(\varphi) \cap U_a = V(g) \cap U_a$$

which is closed in U_a . Since the collection of all such U_a covers U , $V(\varphi)$ is closed in U . $\square$

As desired, we have maintained the restriction property for regular functions. Precisely, if $U \subseteq V \subseteq X$ and U, V are open, then the restriction map $\mathcal{O}_X(V) \rightarrow \mathcal{O}_X(U)$ takes regular functions to regular functions. However, this is not surjective in general. For instance, the element $\varphi(x) = 1/x$ is a regular function in $\mathcal{O}_{\mathbb{A}^1}(\mathbb{A}^1 \setminus \{0\})$ but there is no regular function which restricts to φ from $\mathcal{O}_{\mathbb{A}^1}(\mathbb{A}^1)$. We will later see a theorem that implies this occurs for a broad class of removed singularities. Nevertheless, the restriction is actually injective under the right assumptions.

Proposition 2.2: Identity Theorem

Let X be an irreducible affine variety and let $\emptyset \subsetneq U \subseteq V \subseteq X$ be open. Then if $\varphi_1, \varphi_2 \in \mathcal{O}_X(V)$ and $\varphi_1|_U = \varphi_2|_U$, it follows that $\varphi_1 = \varphi_2$.

Proof. The set $V(\varphi_1 - \varphi_2)$ on which φ_1, φ_2 is closed. It contains U , which is nonempty and open in V . V is irreducible since it is an open subset of X irreducible. So the closure of U in V is V , hence $V \subseteq V(\varphi_1 - \varphi_2)$. Thus $\varphi_1 = \varphi_2$ on V . $\square$

This resembles analytic continuation from complex analysis in spirit. However, it is a less surprising result because we already know that open sets are “large” in the Zariski topology, so it is natural that they extend uniquely.

Now, we compute $\mathcal{O}_X(U)$ for a class of simple open sets.

Definition 2.2

Let X be an affine variety and $f \in A(X)$. Then the **distinguished open subset** for f in X is

$$D(f) = \{x \in X : f(x) \neq 0\}$$

A general open subset is called a distinguished open subset if it is of the form $D(f)$.

Proposition 2.3

The intersection of distinguished open subsets is a distinguished open subset.

Proof. Let $f, g \in A(X)$. Then

$$D(f) \cap D(g) = (X \setminus V_X(f)) \cap (X \setminus V_X(g)) = X \setminus (V_X(f) \cup V_X(g)) = X \setminus V(fg) = D(fg) \quad \square$$

Proposition 2.4

Any open subset in X is a finite union of distinguished subsets.

Proof. Let $U \subseteq X$ be open. Then $U = X \setminus V(I)$ for some ideal $I \subseteq A(X)$, and by Hilbert’s basis theorem we know this can be written as

$$U = X \setminus V_X(\langle f_1, \dots, f_r \rangle)$$

for some finite generators. Then

$$U = X \setminus \bigcap_{i=1}^r V_X(f_i) = \bigcup_{i=1}^r (X \setminus V_X(f_i)) = \bigcup_{i=1}^r D(f_i) \quad \square$$

Thus the distinguished open sets form a basis for the Zariski topology. This is helpful because we showed earlier that being a regular function is a local property. Hence, we can write an open set as a union of distinguished opens, and check if the restriction to each is regular. Thus, we just need to understand $\mathcal{O}_X(D(f))$.

Lemma 2.5: Partition of Unity

Let X be an affine variety and suppose

$$D(f) = \bigcup_{i \in I} D(f_i)$$

Then there is $m \in \mathbb{N}$ such that

$$f^m = \sum_{i \in I_0} r_i f_i$$

where I_0 is a finite subset of I and each $r_i \in A(X)$.

This is basically the Nullstellensatz, so we remark that algebraic closure of the ground field is critical for the existence of partitions of unity. Since each f_i is supported on $D(f_i)$, this breaks f^m into terms which are supported on distinguished open sets. The utility of this is clear from analogy with smooth partitions of unity.

Proof. By the Nullstellensatz,

$$f \in I_X(V_X(f))$$

We want to show that this is the radical of an ideal which is the sum of some principal ideals. Note that

$$V_X(f) = X \setminus D(f) = \bigcap_{i \in I} V_X(f_i) = V_X(\langle f_i : i \in I \rangle)$$

So

$$f \in \sqrt{\langle f_i : i \in I \rangle} \implies f^m = \sum_{i \in I_0} r_i f_i$$

for some finite subset I_0 . □

Functions on distinguished open sets are locally quotients of polynomials except on $V_X(\langle f \rangle)$, which we know has dimension at most 1 by Krull's principal ideal theorem. In some sense, we should expect that this will allow the singularities to act similarly to removable singularities.

Proposition 2.6

For any distinguished open subset $D(f) \subseteq X$, the regular functions on $D(f)$ take the form

$$\mathcal{O}_X(D(f)) = \left\{ \frac{g}{f^m} : g \in A(X), m \in \mathbb{N} \right\}$$

Thus, we see that regular functions on distinguished open subsets are given globally as a polynomial quotient, not just locally. In particular, taking $f = 1$ we get $D(f) = X$ so the regular functions on X are all of $A(X)$.

Proof. ($\supseteq$) This is clear, since any quotient g/f^m makes sense on $D(f)$ and is clearly locally a quotient.

($\subseteq$) Let $\varphi \in \mathcal{O}_X(D(f))$.

Claim 1: for any $a \in D(f)$, φ can be written locally as a quotient

$$\varphi = \frac{g_a}{f_a}$$

on U_a where $D(f_a) = U_a$ and $g_a = 0$ on $V_X(f_a)$.

To see this, first note that we can generally write

$$\varphi = \frac{g_a}{f_a}$$

on U_a for some g_a, f_a with $a \in U_a$. But since the distinguished opens form a basis, there is a distinguished open subset $D(h_a)$ contained in U_a which contains a . On $D(h_a)$,

$$\varphi = \frac{g_a h_a}{f_a h_a}$$

Since $D(h_a) \subseteq D(f_a)$, $V_X(h_a) \supseteq V_X(f_a)$, so h_a vanishes wherever f_a vanishes. So on $V_X(f_a h_a)$, $g_a h_a = 0$. Thus taking $g'_a = g_a h_a$, $f'_a = f_a h_a$, we have that $g'_a = 0$ on $V_X(f'_a)$, proving Claim 1.

Claim 2: For $a, b \in D(f)$ and f_a, g_a, f_b, g_b as constructed above, $g_a f_b = g_b f_a$ as elements of $A(X)$.

To see this, $U_a \cap U_b$ is a nonempty open set, on which φ has two representations

$$\varphi = \frac{g_a}{f_a} = \frac{g_b}{f_b}$$

Clearing denominators, we have $g_a f_b = g_b f_a$ on $U_a \cap U_b$. On the other hand, if $x \notin U_a \cap U_b$, then either $x \in V_X(f_a)$ or $x \in V_X(f_b)$ (since by Claim 1, we chose $U_i = X \setminus V_X(f_i)$). Say $x \in V_X(f_a)$. Then by Claim 1, $g_a(x) = f_a(x) = 0$, so both sides are zero. So everywhere we have $g_a f_b = g_b f_a$, proving Claim 2.

Finally, with g_a, f_a, U_a defined as in Claim 1,

$$\bigcup_{a \in D(f)} U_a = \bigcup_{a \in D(f)} D(f_a) = \bigcup_{a \in D(f)} D(f_a)$$

So by the partition of unity, there is m and a finite subset $A \subseteq D(f)$ such that

$$f^m = \sum_{a \in A} r_a f_a$$

Now, define

$$g = \sum_{a \in A} r_a g_a$$

It follows that for any $x \in D(f)$,

$$gf_x = \sum_{a \in A} r_a g_a f_x \stackrel{(2)}{=} \sum_{a \in A} r_a g_x f_a = g_x \sum_{a \in A} r_a f_a = g_x f^m$$

Since f, f_x are nonzero at x , we have

$$\frac{g(x)}{f^m(x)} = \frac{g_x}{f_x} = \varphi(x)$$

□

We can view this construction in a completely algebraic sense.

Definition 2.3

Let $S \subseteq R$ be a multiplicative subset. Then the **localization** of R at S is the ring of fractions

$$S^{-1}R = R[S^{-1}] = \left\{ \frac{r}{s} : r \in R, s \in S \right\} / \sim$$

where the equivalence relation $\sim$ is such that

$$\frac{r}{s} \sim \frac{r'}{s'}$$

if and only if there is $t \in S$ such that

$$t(rs' - r's) = 0$$

In particular, when S is the subset $S = \{1, f, f^2, \dots\}$ for some $f \in R$, we denote $R_f = R[S^{-1}]$.

Corollary 2.7

Let $f \in A(X)$. Then the ring of regular functions on the distinguished subset $D(f)$ is

$$\mathcal{O}_X(D(f)) \cong A(X)_f$$

which are isomorphic as K -algebras.

Proof. We can consider the homomorphism of K -algebras $\varphi : A(X)_f \mapsto \mathcal{O}_X(D(f))$ given by

$$\varphi \left(\left[\frac{g}{f^m} \right] \right) = \frac{g}{f^m}$$

We check that this is well defined, injective, and surjective. First, if

$$\left[\frac{g}{f^m} \right] = \left[\frac{g'}{f'^m} \right]$$

then there is k such that

$$f^k (gf'^m - g'f^m) = 0$$

Since $f \neq 0$ on $D(f)$, we have $gf'^m - g'f^m = 0$ on $D(f)$, so

$$\frac{g}{f^m} = \frac{g'}{f'^m}$$

as functions on $D(f)$. So this map is well defined. Since $D(f)$ is a distinguished subset, every element of $\mathcal{O}_X(D(f))$ is of the form g/f^m , so this map is surjective. To prove injectivity, if

$$\varphi\left(\left[\frac{g}{f^m}\right]\right) \equiv 0$$

on $D(f)$ then $g = 0$ on $D(f)$, so $fg = 0$ on $D(f)$. Then

$$\left[\frac{g}{f^m}\right] = \left[\frac{0}{1}\right]$$

since $f(g \cdot 1 - f^m \cdot 0) = fg = 0$ on $D(f)$. □

So we have computed the ring of regular functions on distinguished open subsets. As an example of a subset which is not distinguished, we consider $\mathbb{A}^2 \setminus \{(0, 0)\}$.

Example 2.2

We will show that

$$\mathcal{O}_{\mathbb{A}^2}(\mathbb{A}^2 \setminus \{(0, 0)\}) \cong \mathbb{K}[x_1, x_2] \cong \mathcal{O}_{\mathbb{A}^2}(\mathbb{A}^2)$$

which is to say that every regular function on $U = \mathbb{A}^2 \setminus \{(0, 0)\}$ can be extended over the origin. To see this, note that $U = D(x_1) \cup D(x_2) \supseteq D(x_1x_2)$. Letting $\varphi \in \mathcal{O}_{\mathbb{A}^2}(U)$, φ restricts to regular functions on $D(x_i)$, which then restrict to regular functions on $D(x_1x_2)$. This means there are m, n such that

$$\varphi = \frac{g_1}{x_1^m} = \frac{g_2}{x_2^n}$$

for $g_1, g_2 \in K[x_1, x_2]$. The equality holds on $D(x_1x_2)$, so by our characterization of $\mathcal{O}_X(D(x_1x_2))$ we know the two are equal as equivalence classes. Thus there is r such that

$$(x_1x_2)^r(g_1x_2^n - g_2x_1^m) = 0$$

as elements of $K[x_1, x_2]$. Since $K[x_1, x_2]$ is a domain, we have

$$g_1x_2^n = g_2x_1^m$$

Since $K[x_1, x_2]$ is a UFD, we get that x_1^m divides x_2^n or g_1 . By assuming $\gcd(g_1, x_1) = \gcd(g_2, x_2) = 1$ if necessary, which is harmless, we see that $m = 0, n = 0$, so $\varphi = g_1 = g_2$ extends to a polynomial function on $\mathbb{A}^2$.

Proposition 2.8

Let Y be a nonempty irreducible subset of an affine variety X such that $A(X)$ is a UFD. Let $U = X \setminus Y$. Then $\mathcal{O}_X(U) = A(X)$ if and only if $\text{codim}_X(Y) \geq 2$.

Proof. ($\Leftarrow$) Clearly any element of $A(X)$ can be restricted to a regular function on U , so we just show that this is the form of all regular functions. First we show this when $\text{codim}_X Y = 2$. In this case, the height of $I_X(Y)$ in $A(X)$ is 2, so there is a maximal chain of prime ideals

$$(0) \subseteq \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \mathfrak{p}_2 \subseteq I_X(Y)$$

Since Y is irreducible, $I_X(Y)$ is prime, and since $A(X)$ is a domain, (0) is prime. So by maximality this chain is of the form

$$(0) \subsetneq \mathfrak{p} \subsetneq I_X(Y)$$

Now, take any $q \in I_X(Y) \setminus \mathfrak{p}$ and set $\mathfrak{q} = \langle q \rangle$. $\mathfrak{q}$ is not all of $I_X(Y)$, since then $I_X(Y)$ would be principal and would have height at most 1 by Krull's principal ideal theorem. But $I_X(Y)$ has height $\text{codim}_X Y = 2$.

$\sqrt{\mathfrak{p} + \mathfrak{q}}$ is the intersection of prime ideals containing $\mathfrak{p} + \mathfrak{q}$. Any such ideal strictly contains $\mathfrak{p}$, so by maximality of the chain, they also must contain $I_X(Y)$. $I_X(Y)$ is a prime ideal containing $\mathfrak{p} + \mathfrak{q}$, so $\sqrt{\mathfrak{p} + \mathfrak{q}} = I_X(Y)$. Also, $\mathfrak{p} \cap \mathfrak{q} = (0)$, because it is a prime ideal strictly contained in $\mathfrak{p}$ (as $\mathfrak{q}$ is not $I_X(Y)$).

Now, by the inclusion reversing bijection, we have

$$Y \subsetneq V_X(\mathfrak{p}), V_X(\mathfrak{q})$$

where both $V_X(\mathfrak{p}), V_X(\mathfrak{q})$ are irreducible subvarieties of X such that

$$V_X(\mathfrak{p}) \cap V_X(\mathfrak{q}) = V_X(\sqrt{\mathfrak{p} + \mathfrak{q}}) = V_X(I_X(Y)) = Y$$

$\mathfrak{p}, \mathfrak{q}$ are both finitely generated, so let

$$\begin{aligned} \mathfrak{p} &= \langle p_1, \dots, p_n \rangle \subseteq A(X) \\ \mathfrak{q} &= \langle q_1, \dots, q_m \rangle \subseteq A(X) \end{aligned}$$

In particular, the requirement that $\mathfrak{p} \cap \mathfrak{q} = (0)$ guarantees that $p_i \nmid q_j$ and $q_j \nmid p_i$ for any i, j . We have

$$U = X \setminus Y = X \setminus (V_X(\mathfrak{p}) \cap V_X(\mathfrak{q})) = (X \setminus V_X(\mathfrak{p})) \cup (X \setminus V_X(\mathfrak{q})) \supsetneq D(p_1 \cdots p_n) \cup D(q_1 \cdots q_m)$$

Write $D_p = D(p_1 \cdots p_n)$, $D_q = D(q_1 \cdots q_m)$, $D = D_p \cap D_q$. D is nonempty since it is the intersection of nonempty open sets.

Pick $\varphi \in \mathcal{O}_X(U)$. Then $\varphi|_{D_p}, \varphi|_{D_q}$ are both regular functions on distinguished open subsets, so there are $k, l \in \mathbb{N}, g_p, g_q \in A(X)$ such that

$$\varphi|_{D_p} = \frac{g_p}{(p_1 \cdots p_n)^k}, \quad \varphi|_{D_q} = \frac{g_q}{(q_1 \cdots q_m)^l}$$

(We assume that $\gcd(g_p, p_i) = \gcd(g_q, q_j) = 1$, which is harmless.) Both representations are equal to φ on D , so on D we have

$$g_p(q_1 \cdots q_m)^l = g_q(p_1 \cdots p_n)^k \in A(X)$$

The locus for this equality is closed, so in particular it contains $\overline{D}$.

(since D is nonempty and open, and X is irreducible). The p_i are coprime with the q_j , and we assumed that g_p was coprime with the p_i and g_q coprime with the q_j . So since $A(X)$ is a UFD, we have $l = k = 0$, and thus $\varphi = g_p = g_q$ on D . Since this is an agreement of regular functions on an open set, by the identity theorem we have that $\varphi = g_p = g_q$ on all of U . So $\mathcal{O}_X(U) = A(X)$ when $\text{codim}_X Y = 2$.

To see the general case, let $\text{codim}_X Y \geq 2$. Then since $\dim X$ is finite, there is $Y' \supseteq Y$ with $\text{codim}_X Y' = 2$. Let $\varphi \in \mathcal{O}_X(U)$. Then φ restricts to a regular function on $X \setminus Y' \subseteq U$. By the theorem for codimension 2, we have

$$\varphi|_{X \setminus Y'} = g|_{X \setminus Y'}$$

for some $g \in A(X)$. Since φ, g agree on an open subset of U , $\varphi = g$ on U as well. So $\mathcal{O}_X(U) = A(X)$ in the general case.

($\implies$) Now suppose that $\mathcal{O}_X(U) = A(X)$. Clearly we cannot have $\text{codim}_X Y = 0$, since then $Y = X$ and $\mathcal{O}_X(U)$ is the zero ring. So all that we need to show is that $\text{codim}_X Y \neq 1$.

Suppose for contradiction that $\text{codim}_X Y = 1$. $\text{codim}_X Y$ is the height of $I_X(Y)$ in $A(X)$, so $I_X(Y)$ has height 1. It is prime since Y is irreducible. Then since $A(X)$ is a Noetherian UFD, every prime ideal of height 1 is principal (for instance, Gathmann Proposition 2.37). So we have $I_X(Y) = \langle f \rangle$ for some nonunit $f \in A(X)$. Then $1/f \in \mathcal{O}_X(U)$ is a valid representation of a regular function. But if $1/f = g \in \mathcal{O}_X(U)$ for some $g \in A(X)$, then we would have

$$1 = fg$$

on U . Then $V(1 - fg) \supseteq \overline{U} = X$, so $fg = 1 \in A(X)$. But f is not a unit, contradiction. Thus $\text{codim}_X Y \geq 2$. $\square$

2.2 Sheaves

We defined regular functions in such a way that:

1. Restrictions of regular functions are regular,
2. Regularity can be checked locally, meaning that if a function is regular on each open set in an open cover, then it is a regular function everywhere,
3. Constructing regular functions by “gluing” together functions on open subsets is unique (by the identity theorem).

The notion of a sheaf is an abstract view of how data of this type can be collected into a single object. We will formalize this so that we can discuss mappings from an affine variety into another one (other than $\mathbb{A}^1$, which we have just developed), seeing affine varieties as not simply a set in $\mathbb{A}^n$ but as such a set together with the data given by the sheaf.

Definition 2.4

Let (X, τ) be a topological space. Then a **presheaf** of rings on X is a collection of rings $\{\mathcal{F}(U)\}_{U \in \tau}$, together with ring homomorphisms $\rho_{V \rightarrow U} : \mathcal{F}(V) \rightarrow \mathcal{F}(U)$ for each inclusion of opens $U \subseteq V$. We also require that $\rho_{V \rightarrow V}$ is the identity on $\mathcal{F}(V)$, and that whenever $U \subseteq V \subseteq W$, the following diagram commutes:

$$\begin{array}{ccccc} \mathcal{F}(U) & \xrightarrow{\rho_{U \rightarrow V}} & \mathcal{F}(V) & \xrightarrow{\rho_{V \rightarrow W}} & \mathcal{F}(W) \\ & \searrow \rho_{U \rightarrow W} & & & \end{array}$$

Elements of $\mathcal{F}(V)$ are called **sections** of $\mathcal{F}$ on V , and elements of $\mathcal{F}(X)$ are called **global sections**. The maps $\rho_{V \rightarrow U}$ are also called **restriction maps**, and $\rho_{V \rightarrow U}(\varphi)$ is denoted $\varphi|_U$.

$\mathcal{F}(U)$ should intuitively be thought of as a ring of functions with the restriction maps as actual restrictions, but we do not require this.

Definition 2.5

A presheaf $\mathcal{F}$ is called a **sheaf** if it satisfies the gluing properties:

1. Whenever $\{U_i\}_{i \in I}$ is an open cover of $U \subseteq X$ open, if $\varphi_i \in \mathcal{F}(U_i)$ for each i and $\varphi_i|_{U_i \cap U_j} = \varphi_j|_{U_i \cap U_j} \in \mathcal{F}(U_i \cap U_j)$ for each i, j , then there is $\varphi \in \mathcal{F}(U)$ such that $\varphi|_{U_i} = \varphi_i$,
2. Such a φ is unique.

In essence, compatible data can be glued together on small pieces to produce larger data.

We also note that this makes sense in many categories (basically any reasonable category which is a set with structure). For instance, we can have a sheaf of K -algebras, R -modules, abelian groups, and so on.

Example 2.3

The continuous ring-valued functions on a topological space form a sheaf of rings. If they are K -valued, then it is also a sheaf of K -algebras. The differentiable functions on $\mathbb{R}^d$ also form a sheaf of K -algebras.

Example 2.4

If X is any space and R a ring, then the constant functions $f : U \rightarrow R$, $U \subseteq X$ form the **constant presheaf**. However, this is in general not a sheaf, because if we can take two disjoint open sets, then we can prescribe different values on each. This is compatible data since the intersection is empty, but it does not glue into a constant function. Intuitively, being constant is not a local property. The *locally* constant functions do form a sheaf, though.

Proposition 2.9

If X is an affine variety, then the sheaf of regular functions $\mathcal{F}(U) = \mathcal{O}_X(U)$ with the restriction maps given by function restriction form a sheaf.

Proof. That this is a presheaf is clear. Since being regular is a local property, compatible regular functions can be glued into regular functions, and by the identity theorem they are unique. $\square$

Definition 2.6

Let $W \subseteq X$ be open and $\mathcal{F}$ a presheaf (resp. sheaf) on X . Then the restriction presheaf (sheaf) on W is

$$\mathcal{F}|_W(U) = \mathcal{F}(U)$$

for $U \subseteq W$ open, with the same restriction maps as in $\mathcal{F}$.

If sheaves of functions collect data of functions, then we can look at the behavior of functions around a given point in the topological space. The possible behaviors of these functions are described by stalks.

Definition 2.7

Let $\mathcal{F}$ be a presheaf over X and $a \in X$. Then the **stalk** of $\mathcal{F}$ at a is

$$\mathcal{F}_a = \{(U, \varphi) : \varphi \in \mathcal{F}(U)\} / \sim$$

with the equivalence relation

$$(U, \varphi) \sim (V, \psi)$$

if and only if there is $W \subseteq U \cap V$ open containing a such that $\varphi|_W = \psi|_W$. An element of $\mathcal{F}_a$ is called a **germ** at a , or sometimes a **local section**.

A stalk has the structure of a ring if $\mathcal{F}$ is a presheaf of rings, of K -algebras if $\mathcal{F}$ is a presheaf of K -algebras, and so on. For instance, we have

$$\begin{aligned} [(U_1, \varphi_1)] + [(U_2, \varphi_2)] &= [(U_1 \cap U_2, \varphi_1 + \varphi_2)] \\ [(U_1, \varphi_1)] \cdot [(U_2, \varphi_2)] &= [(U_1 \cap U_2, \varphi_1 \varphi_2)] \end{aligned}$$

When we work with the sheaf of regular functions, we denote this $\mathcal{O}_{X,a}$. The information contained in a germ or stalk is the local behavior of a function. The strength of this information is dependent on the specific sheaf or presheaf.

Example 2.5

Consider the sheaf of continuous functions on X and consider a germ $\varphi_a \in \mathcal{F}_a$ for $a \in X$. Then every representative of the germ agrees when evaluated at a , so $\varphi_a(a)$

is well defined (though $\varphi_a(a)$ does not uniquely determine a germ). However, a germ cannot be evaluated anywhere else, since there are continuous functions in the same germ with different values there.

Example 2.6

Consider the sheaf of holomorphic functions on $\mathbb{C}$ and consider a germ $\varphi_a \in \mathcal{F}_a$. Then any two representatives agree on an open set containing a . Up to ignoring representatives defined on disconnected sets, regularity tells us that each germ corresponds to exactly one holomorphic function. Thus any section on, say, $\mathbb{D}$, is entirely determined by its germ at 0, and the germ can be evaluated anywhere on its maximal domain.

Example 2.7

For the sheaf of regular functions on an affine variety, there is a well-defined evaluation map $\varphi_a \mapsto \varphi(a)$.

We will now see how to compute the stalks of regular functions over affine varieties. This algebraic operation is called localization, because it reflects the fact that we are looking at the properties of these functions locally.

Definition 2.8

Let R be a ring and $\mathfrak{p} \subseteq R$ an ideal. Let $S_{\mathfrak{p}} = R \setminus \mathfrak{p}$ (which is multiplicative). Then the **localization** of R around $\mathfrak{p}$ to be the localization $R_{\mathfrak{p}} = R[S_{\mathfrak{p}}^{-1}]$.

Lemma 2.10

Let $\pi : R \rightarrow R_{\mathfrak{p}}$ be the natural inclusion map $r \mapsto \left[\frac{r}{1} \right]$. Then the maps

$$\{\mathfrak{q} \subseteq R_{\mathfrak{p}} \text{ prime}\} \xrightleftharpoons[\mathfrak{q}' \cdot R_{\mathfrak{p}} \leftarrow \mathfrak{q}']{\mathfrak{q} \mapsto \pi^{-1}(\mathfrak{q})} \{\mathfrak{q}' \subseteq R \text{ prime, contained in } \mathfrak{p}\}$$

are inclusion preserving bijections which are inverses.

Definition 2.9

A **local ring** is a ring with a unique maximal ideal.

Corollary 2.11

If $\mathfrak{p}$ is prime then $R_{\mathfrak{p}}$ is a local ring with maximal ideal $\mathfrak{p} \cdot R_{\mathfrak{p}}$.

Proof. The unique prime ideal contained in $\mathfrak{p}$ which is maximal among such ideals is $\mathfrak{p}$ itself. Passing through the bijection, the unique maximal ideal in $R_{\mathfrak{p}}$ is $\mathfrak{p} \cdot R_{\mathfrak{p}}$. $\square$

Lemma 2.12

Let $a \in X$ be a point in an affine variety. The map from the localization of the coordinate ring to the stalk at a

$$A(X)_{I_X(a)} \rightarrow \mathcal{O}_{X,a}$$

given by

$$\left[\frac{g}{f} \right]_{A(X)_{I_X(a)}} \mapsto \left[\left(D(f), \frac{g}{f} \right) \right]_{\mathcal{O}_{X,a}}$$

is an isomorphism of K -algebras.

Proof. We check that this map is well defined, surjective, and injective. Being a K -algebra morphism is clear.

To see that it is well defined, if

$$\left[\frac{g}{f} \right]_{A(X)_{I_X(a)}} = \left[\frac{g'}{f'} \right]_{A(X)_{I_X(a)}}$$

then there is $h \notin I_X(a)$ such that

$$h(fg' - gf') = 0 \in A(X)$$

So on

$$D(h) \cap D(f) \cap D(f')$$

we have

$$\frac{g}{f} = \frac{g'}{f'}$$

pointwise. This is a subset of both $D(f), D(f')$, so

$$\left[\left(D(f), \frac{g}{f} \right) \right]_{\mathcal{O}_{X,a}} = \left[\left(D(f'), \frac{g'}{f'} \right) \right]_{\mathcal{O}_{X,a}}$$

For surjectivity, pick

$$[(U, \varphi)]_{\mathcal{O}_{X,a}}$$

Then φ is regular so there is U_a open containing a and $g, f \in A(X)$ such that $\varphi = g/f$ on U_a and $f \neq 0$ on U_a . In particular we can assume that $U_a = D(f)$. Then we have that

$$\varphi = \frac{g}{f}$$

on $a \in U \cap D(f)$ nonempty. Thus we have that

$$[(U, \varphi)]_{\mathcal{O}_{X,a}} = \left[\left(D(f), \frac{g}{f} \right) \right]_{\mathcal{O}_{X,a}}$$

So this is the image of

$$\left[\frac{g}{f} \right]_{A(X)_{I_X(a)}}$$

To check injectivity, if

$$\left[\left(D(f), \frac{g}{f} \right) \right]_{\mathcal{O}_{X,a}} = \left[\left(X, \frac{0}{1} \right) \right]_{\mathcal{O}_{X,a}}$$

Then there is $U \subseteq D(f)$ containing A such that

$$\frac{g}{f} \Big|_U = 0 \in \mathcal{O}_X(U)$$

Since the distinguished opens form a basis for the Zariski topology, there is h such that $D(h) \subseteq U$. In particular, we must have $h \in A(X) \setminus I_X(a)$. Then

$$\frac{g}{f} \Big|_{D(h)} = \frac{0}{1} \in \mathcal{O}_X(D(h)) = A(X)_h$$

Therefore by definition, there is m such that

$$h^m(g \cdot 1 - f \cdot 0) = h^m g = 0 \in A(X)$$

Since $h \notin I_X(a)$,

$$\left[\frac{g}{f} \right]_{A(X)_{I_X(a)}} = \left[\frac{0}{1} \right]_{A(X)_{I_X(a)}}$$

□

Corollary 2.13

$\mathcal{O}_{X,a}$ is a local ring and the unique maximal ideal is given by

$$I_X(a)A(X)_{I_X(a)} = \left\{ \frac{g}{f} : f \notin I_X(a), g \in I_X(a) \right\} / \sim = \left\{ \frac{g}{f} : f(a) \neq 0, f(a) = 0 \right\} / \sim$$

where $\sim$ is as for localizations.

Proof. $I_X(a)$ is prime. □

Definition 2.10

The **local ring** of X at a is $\mathcal{O}_{X,a}$ and its maximal ideal is denoted

$$I_a := I_X(a)A(X)_{I_X(a)} = \left\{ \frac{g}{f} : f(a) \neq 0, g(a) = 0 \right\}_{/\sim}$$

2.3 Ringed Spaces and Morphisms

Now that we have a characterization of nice functions on affine varieties, we can see morphisms between them as (continuous) maps which preserve those functions. That is, we want a function to have the property that whenever φ is a regular function on U , then $\varphi \circ f$ is a regular function on $f^{-1}(U)$.

$$X \xrightarrow{f} Y$$

$$\mathcal{O}_X(f^{-1}(U)) \xleftarrow{f^*} \mathcal{O}_Y(U)$$

Definition 2.11

A **ringed space** is a topological space with a sheaf of rings $\mathcal{O}_X$ on X . The sheaf is called the **structure sheaf**.

In the case that X is an affine variety we always take the structure sheaf to be the sheaf of regular functions. Any open subset of X is also a ringed space using the restricted sheaf $\mathcal{O}|_U$. For general rings, the composition $\varphi \circ f$ does not necessarily make sense. As a result, we adopt the convention from here until the introduction of schemes that for any ringed space, $\mathcal{F}(U)$ is always a subset of the set of functions from $U \rightarrow K$.

Definition 2.12

Let X, Y be ringed spaces and $f : X \rightarrow Y$ any map. Then for any $\varphi : U \rightarrow K$ for $U \subseteq Y$ open, the map $\varphi \circ f : f^{-1}(U) \rightarrow K$ is called the **pullback** of φ by f and is denoted $f^*\varphi$. If f is continuous and for any $U \subseteq Y$ open, $\varphi \in \mathcal{O}_Y(U)$, we have $f^*\varphi \in \mathcal{O}_X(f^{-1}(U))$, then f is called a **morphism** of ringed spaces. f is an isomorphism of ringed spaces if it is bijective and f^{-1} is also a morphism. We say that a map $f : X \rightarrow Y$ continuous for X, Y affine varieties is a **morphism** of affine varieties if it is a morphism as a ringed space.

In the case that f is a morphism of ringed spaces, then $f^* : \mathcal{O}_Y(U) \rightarrow \mathcal{O}_X(f^{-1}(U))$ is a K -algebra morphism for each U .

Proposition 2.14

The identity map on a ringed space X is a morphism, and if $f : X \rightarrow Y$, $g : Y \rightarrow Z$ are morphisms of ringed spaces, then $g \circ f : X \rightarrow Z$ is a morphism as well.

This tells us that the ringed spaces form a category.

Proposition 2.15

If $f : X \rightarrow Y$ is a morphism of ringed spaces and $U \subseteq X, V \subseteq Y$ are open with $V \subseteq f(U)$, then $f|_U : U \rightarrow V$ is a morphism of ringed spaces.

In particular, the inclusion map $U \rightarrow X$ is a restriction of the identity, so a morphism.

Lemma 2.16

Let $f : X \rightarrow Y$ be a map of ringed spaces, and let $\{U_i\}_{i \in I}$ be an open cover of X such that all the restrictions $f|_{U_i} : U_i \rightarrow Y$ are morphisms of ringed spaces. Then f is a morphism.

Proof. f is continuous since it is continuous on an open cover. Now, let $V \subseteq Y$ be open, and let $\varphi \in \mathcal{O}_Y(V)$. Then for each i ,

$$(f^*\varphi)|_{U_i \cap f^{-1}(V)} = (f|_{U_i \cap f^{-1}(V)})^* \varphi \in \mathcal{O}_X(U_i \cap f^{-1}(V))$$

by assumption. So the sets $\{U_i \cap f^{-1}(V)\}_{i \in I}$ are an open cover of $f^{-1}(V)$, on which the sections $f^*\varphi|_{U_i \cap f^{-1}(V)}$ agree on intersections. So by the gluing property there is a unique map which agrees with the restrictions, which must be $f^*\varphi$. So $f^*\varphi$ is a section in $\mathcal{O}_X(f^{-1}(V))$, and therefore f is a morphism. $\square$

2.4 Morphisms of Affine Varieties

We have thus defined morphisms of ringed spaces, and defined morphisms of affine varieties to be morphisms as ringed spaces. But because we have studied the structure sheaf on affine varieties, we can make more concrete statements about their morphisms.

Proposition 2.17

If $U \subseteq X$ is an open subset of an affine variety and $Y \subseteq \mathbb{A}^n$ an affine variety. Then the morphisms $f : U \rightarrow Y$ are all the maps of the form

$$f(x) = (\varphi_1(x), \dots, \varphi_n(x))$$

where $\varphi_i \in \mathcal{O}_X(U)$ for each i and $f(U) \subseteq Y$.

Proof. Let $f : U \rightarrow Y$ be a morphism. The coordinate projection map $\pi_i : x \mapsto x_i$ on $\mathbb{A}^n$ is a regular function (hence on Y by restriction), so for each i we have that

$$f^* \pi_i(x) = \pi_i(f(x)) \in \mathcal{O}_X(U)$$

by the definition of a morphism. So every morphism is of the above form.

Now, let (φ_i) be regular functions in $\mathcal{O}_X(U)$. We need to show that

$$f = (\varphi_1, \dots, \varphi_n)$$

is a morphism whenever $f(U) \subseteq Y$. Let $Z \subseteq Y$ be closed, so that $Z = V_Y(g_1, \dots, g_m)$ for $g_j \in A(Y)$. Then it follows that

$$f^{-1}(Z) = \{x \in X : \forall j, g_j(\varphi_1, \dots, \varphi_n) = 0\}$$

Since g_j is a polynomial and φ_i lie in $\mathcal{O}_X(U)$, which is a K -algebra, we have that

$$g_j(\varphi_1, \dots, \varphi_n) = f^* g_j$$

is also a regular function, and

$$f^{-1}(Z) = V(f^* g_1) \cap \dots \cap V(f^* g_m)$$

is closed in U (each $V_X(f^* g_i)$ is closed since it is the zero locus of a regular function). Now, let $\psi \in \mathcal{O}_Y(W)$ be a regular function. It follows that locally it is a quotient of polynomials, so

$$f^* \psi = \frac{g(\varphi_1, \dots, \varphi_n)}{h(\varphi_1, \dots, \varphi_n)}$$

is locally a quotient of polynomials, after expanding. Thus $f^* \psi$ is regular, so f is a morphism. $\square$

Example 2.8

Let $X = V(xy - 1) \in \mathbb{A}^2$ and $Y = \mathbb{A}^1$. Then the map $(x, y) \mapsto x : X \rightarrow Y$ is a morphism. If we let $U = \mathbb{A}^1 \setminus \{0\} \subseteq Y$, then this map is surjective, and the map $x \mapsto (x, 1/x)$ is also a morphism which is its inverse. Thus $X \cong U$ as ringed spaces.

Corollary 2.18

Let X, Y be affine varieties. Then there is a 1-1 correspondence between morphisms $X \rightarrow Y$ and K -algebra homomorphisms $A(Y) \rightarrow A(X)$.

Note the contravariance, since we are looking at the pullback.

Proof. Let $f : X \rightarrow Y$ be a morphism. Then by definition of the pullback, $f^* : \mathcal{O}_Y(Y) \rightarrow \mathcal{O}_X(X)$ is a K -algebra homomorphism:

$$f^*(\varphi + \psi) = (\varphi + \psi)(f(\cdot)) = f^* \varphi + f^* \psi$$

and so on. These rings are precisely $A(Y), A(X)$.

On the other hand, let $g : A(Y) \rightarrow A(X)$. Let $y_1, \dots, y_n$ be the coordinate functions on $Y \subseteq \mathbb{A}^n$, and define $\varphi_i = gy_i \in A(X)$. Then let

$$f = (\varphi_1, \dots, \varphi_n)$$

which is a priori a set-theoretic map from $X \rightarrow \mathbb{A}^n$. To see that this is a morphism, for $h \in K[y_1, \dots, y_n]$,

$$(f^*h)(x) = h(\varphi_1(x), \dots, \varphi_n(x)) = h(gy_1(x), \dots, gy_n(x)) = g(h(y_1, \dots, y_n))(x)$$

where the last equality follows because K -algebra homomorphisms commute with polynomials. Now, for any $h \in I(Y)$, and $x \in X$, we have

$$h(f(x)) = f^*h(x) = (g(h(y)))(x)$$

since $h \in I(Y)$, $h = 0 \in A(Y)$, so $h(f(x)) = 0$. Thus $f(X) \subseteq V(I(Y)) = Y$, so the map is indeed into Y . To see that it is a morphism, we just note that the coordinate functions φ_i are regular, so by the previous theorem f is a morphism. Then we finally see that $f^* = g$, so we have constructed the morphism corresponding to g . $\square$

Corollary 2.19

Two affine varieties are isomorphic if and only if their coordinate rings are isomorphic.

Example 2.9

Let X be the rational normal curve $V(y - x^2, z - x^3) \subseteq \mathbb{A}^3$. Then

$$A(X) = \frac{K[x, y, z]}{\langle y - x^2, z - x^3 \rangle} = K[x]$$

so $X \cong \mathbb{A}^1$.

Proposition 2.20

The automorphism group of $\mathbb{A}^1$ is

$$\text{Aut}(\mathbb{A}^1) = \{f(x) = ax + b, a \in K^*, b \in K\}$$

Proof. The K -algebra homomorphisms of $K[x]$ are those of the form

$$\psi_p : f \mapsto f(p(x))$$

But $\deg f \circ p \geq \deg f \deg p$, and when f is a constant function $f \circ p$ is also constant. Thus if p has degree higher than 1 there are no degree 1 polynomials in the image, and if p is constant then so is ψ_p . Thus the only isomorphisms are of the form $\psi_p, p = ax + b$ where $a \neq 0$. From

correspondence with K -algebra homomorphisms we know that the corresponding morphism is

$$f = (p(x_1), \dots, p(x_n))$$

but we work over $n = 1$, so this is just $f = p$. $\square$

Example 2.10

Let $X = V(x^2 - y^3) \subseteq \mathbb{A}^2$. Then $f : \mathbb{A}^1 \rightarrow X$ given by

$$t \mapsto (t^3, t^2)$$

is bijective and a morphism (since its coordinates are regular). But the pullback map $f^* : K[x, y]/\langle x^2 - y^3 \rangle \rightarrow K[t]$ maps

$$[x] \mapsto [t^3], [y] \mapsto [t^2]$$

so there are no degree 1 elements in the image, hence this is not an isomorphism. So it is not enough to be a bijective morphism to be an isomorphism.

We can use this correspondence and the categorical interpretation of the product to give a categorical characterization of the coordinate ring of products.

Proposition 2.21

If X, Y are affine varieties then the projection map

$$\pi_X : (x_1, \dots, x_n, y_1, \dots, y_m) \mapsto (x_1, \dots, x_n)$$

is a morphism, and similarly for Y .

Proposition 2.22: Universal Property of Products

Let X, Y be affine varieties and π_X, π_Y be projections from $X \times Y \rightarrow X, Y$. If Z is an affine variety and $f_X : Z \rightarrow X, f_Y : Z \rightarrow Y$ are maps then there is a unique morphism $Z \rightarrow X \times Y$ that makes the diagram commute:

$$\begin{array}{ccccc} Z & & & & \\ & \searrow f_Y & & \searrow & \\ & & X \times Y & \xrightarrow{\pi_Y} & Y \\ & \swarrow f_X & \downarrow \pi_X & & \\ & & X & & \end{array}$$

$\exists! f$ (dashed arrow from Z to $X \times Y$)

Proof. The only set theoretic map that satisfies this is $f = (f_X, f_Y)$ by looking at the projections of f . This is a morphism since the coordinates of f_X, f_Y are regular, so so are the coordinates of f . $\square$

By passing through the contravariant correspondence, we get the universal property of tensor products on the coordinate rings.

$$\begin{array}{ccc}
 \begin{array}{c}
 Z \xrightarrow{f_Y} Y \\
 \searrow \exists! f \\
 X \times Y \xrightarrow{\pi_Y} Y \\
 \downarrow \pi_X \\
 X
 \end{array}
 & \Longleftrightarrow &
 \begin{array}{c}
 A(Z) \xleftarrow{g_Y} A(Y) \\
 \nwarrow \exists! g \\
 A(X) \otimes_K A(Y) \xleftarrow{\pi_Y^*} A(Y) \\
 \uparrow \pi_X^* \\
 A(X)
 \end{array}
 \end{array}$$

We have seen that every affine variety may be associated with its coordinate ring $A(X)$, which is a finitely generated, reduced K -algebra. Moreover, if we restrict to equivalence classes, then this map injects the equivalence classes of affine varieties into the finitely generated reduced K -algebras. We see the reverse of this correspondence now.

Let R be a f.g. reduced K -algebra. Then let $b_1, \dots, b_n$ generate R . This gives a surjective K -algebra homomorphism $g : K[x_1, \dots, x_n] \rightarrow R$ defined by

$$g(x_i) = b_i$$

or equivalently $f \mapsto f(b_1, \dots, b_n) \in R$. By the isomorphism theorem, we have

$$R \cong K[x_1, \dots, x_n] / \ker g$$

Set $I = \ker g$. Since R is reduced, I is radical. So if we take $X = V(I) \subseteq \mathbb{A}^n$, then

$$A(X) = K[x_1, \dots, x_n] / I(X) = K[x_1, \dots, x_n] / I(V(I)) = K[x_1, \dots, x_n] / I \cong R$$

This is not a unique representation of R , but any two representations are isomorphic as affine varieties. Thus we see that the contravariant functor from affine varieties to finitely generated reduced K -algebras is an equivalence of categories, meaning that it is bijective on equivalence classes of objects such that $\text{Mor}(X, Y) \cong \text{Mor}(\mathcal{O}_Y(Y), \mathcal{O}_X(X))$. Moreover, since from a categorical perspective, affine varieties are simply specific ringed spaces, we can extend our definitions:

Remark 2.1

From now, an affine variety is *any* ringed space isomorphic to an affine variety (in the old sense) as a ringed space. We write $A(X) = \mathcal{O}_X(X)$, where $\mathcal{O}_X$ is the structure sheaf.

This allows us to see certain open subsets of affine varieties as affine varieties (though they are not subvarieties in the old sense).

Proposition 2.23

Let X be an affine variety and let $f \in A(X)$. Then $D(f)$ is an affine variety with coordinate ring

$$A(D(f)) = \mathcal{O}_X(D(f)) = A(X)_f$$

Proof. Write

$$Y = \{(x, t) \in X \times \mathbb{A}^1 : tf(x) - 1 = 0\} \subseteq X \times \mathbb{A}^1$$

There is a projection morphism $g : Y \rightarrow D(f)$ mapping $(x, t) \mapsto x$, since $f(x)$ is always nonzero for elements of Y . The inverse is given by

$$x \mapsto \left(x, \frac{1}{f(x)}\right)$$

$\frac{1}{f}$ is an element of $\mathcal{O}_X(D(f))$, so this is a morphism and it is the inverse of g . So $D(f)$ is an affine variety as a sub ringed space, meaning that

$$A(D(f)) \cong \mathcal{O}_X(D(f))$$

and we know that

$$\mathcal{O}_X(D(f)) \cong A(X)_f$$

□

Example 2.11

$U = \mathbb{A}^2 \setminus \{(0, 0)\}$ is an open subset. Suppose it were an affine variety. Then its coordinate ring would be $\mathcal{O}_{\mathbb{A}^2}(U) = K[x, y]$, and the identity map on $K[x, y]$ induces the isomorphism

$$\text{id} : U \rightarrow \mathbb{A}^2$$

but this is not a bijection, hence not an isomorphism, contradiction. Thus U is not an affine variety at all.

2.5 Varieties

We conclude the development of affine varieties by generalizing our results to spaces that only look locally like an affine variety. For instance, we showed in the last section that $U = \mathbb{A}^2 \setminus \{(0, 0)\}$ is not an affine variety. However, it is the union $D(x) \cup D(y)$, so every point has a neighborhood which is an affine variety. Thus we allow for more general constructions where affine varieties are glued together into larger spaces, analogously to how manifolds are glued copies of Euclidean space.

This will allow us to remedy some issues with the more restrictive definition of affine varieties. For instance, in the Euclidean topology, no affine variety is compact except the finite sets (they are all compact in the Zariski topology, or more accurately quasi-compact since this is not Hausdorff). Moreover, arbitrary open subsets of affine varieties are not varieties, as we saw above.

Definition 2.13

A **prevariety** is a ringed space X which admits a finite open cover by affine varieties. A morphism of a prevariety is a morphism as a ringed space. The regular functions on $U \subseteq X$ open are the elements of $\mathcal{O}_X(U)$. An open subset of a prevariety which is an affine variety is an affine open set.

Example 2.12

$U = \mathbb{A}^2 \setminus \{(0, 0)\}$ is a prevariety covered by $D(x), D(y)$, which are distinguished open subsets, hence affine varieties. More generally, since the distinguished open subsets form a basis, any open subset of an affine variety has a finite cover by distinguished open subsets, so is a prevariety.

Proposition 2.24

Any open subset of a prevariety is a prevariety.

Proof. Let X be a prevariety covered by $(U_i)_{i=1}^n$ affine varieties, and let $Y \subseteq X$ be open. Then $(Y \cap U_i)_{i=1}^n$ is an open cover of Y . Moreover, each $Y \cap U_i$ is an open subset of U_i , so it is an open subset of an affine variety, hence a prevariety. Thus there are $(V_{i,j})_{j=1}^{m_n}$ affine varieties covering $Y \cap U_i$, and taking all $V_{i,j}$, we have a finite open cover of Y by affine varieties. $\square$

Since the point of prevarieties is to allow for patching together affine varieties locally, we should develop a tool to do this. We will begin by treating the case of two prevarieties.

Definition 2.14

Let X_1, X_2 be two prevarieties and let $U_1 \subseteq X_1, U_2 \subseteq X_2$ be open subsets, with an isomorphism of prevarieties $f : U_1 \rightarrow U_2$. Then the prevariety X formed by gluing X_1, X_2 along f is the ringed space $(X_1 \sqcup X_2) / \sim$, where $\sim$ is the equivalence relation $x \sim x$, together with $a \sim f(a), f(a) \sim a$ for $a \in U_1$. This set is endowed with the quotient topology, and the structure sheaf is given by

$$\mathcal{O}_X(U) = \{\varphi : U \rightarrow K : i_1^* \varphi \in \mathcal{O}_{X_1}(i_1^{-1}(U)), i_2^* \varphi \in \mathcal{O}_{X_2}(i_2^{-1}(U))\}$$

where $i_1 : X_1 \rightarrow X, i_2 : X_2 \rightarrow X$ are the injection maps which take x to its equivalence class in X .

Since X_1, X_2 are prevarieties isomorphic to their images, we get a finite cover of the glued space by affine varieties, so the glued space is indeed a prevariety. Also, it is immediate that $\mathcal{O}_X|_{X_i} = \mathcal{O}_{X_i}$ for $i = 1, 2$.

Example 2.13

Let $X_1 = X_2 = \mathbb{A}^1$, with $U_1 = U_2 = \mathbb{A}^1 \setminus \{0\}$. If $f(x) = 1/x$, then the glued space is the space $\mathbb{A}^1$ together with a point at infinity. This is called the **projective line** $\mathbb{P}^1$. In particular when $K = \mathbb{C}$, this is the Riemann sphere. If we consider the morphisms $X_1 \rightarrow X_2 : x \mapsto x$ and $X_2 \rightarrow X_1 : x \mapsto x$, this maps $[x] \mapsto [1/x]$ for $x \in X_1$, and similarly for X_2 . Moreover, these agree on their combined domain, which is the image of U_1 (and U_2 , by the quotient equivalence relation). Thus the glued map is a morphism on $\mathbb{P}^1 \rightarrow \mathbb{P}^1$, which takes $x \mapsto 1/x$ for $x \in K^*$ and $0 \leftrightarrow \infty$. On the other hand, $f(x) = x$, we get a space that looks like $\mathbb{A}^1$ but with two points at the origin. This is a non-Hausdorff space over $\mathbb{C}$, since we can take a sequence which converges to both origins. The glued morphism as above is the identity on $\mathbb{A}^1 \setminus \{0\}$ but switches the origins. As a result, the set $\{x : g(x) = x\}$ is the equality locus of two morphisms, but is not a closed set in X .

Pathological behavior like the affine line with two origins will eventually be excluded when we move from prevarieties to varieties. The extension to finite collections of prevarieties is straightforward and just requires a few extra compatibility conditions.

Definition 2.15

Let $(X_i)_{i=1}^n$ be a finite set of prevarieties. For all $i \neq j$ suppose there are $U_{ij} \subseteq X_i$ open and an isomorphism $f_{ij} : U_{ij} \rightarrow U_{ji}$ such that whenever i, j, k are distinct,

$$f_{ij}^{-1} = f_{ji}$$

and $f_{ij}^{-1}(U_{jk}) \subseteq U_{ik}$ with

$$f_{jk} \circ f_{ij} = f_{ik}$$

on $f_{ij}^{-1}(U_{jk})$. Then we define the glued prevariety to be $X = \sqcup X_i / \sim$, where the equivalence relation $\sim$ is defined by $x \sim x$ with $a \sim f(a)$, $f(a) \sim a$ for all $a \in U_{ij}$. By the compatibility relations, $\sim$ is transitive and symmetric, so it is an equivalence relation, and the structure sheaf and topology are defined the same as above.

Example 2.14

Let $X_0 = X_1 = X_2 = \mathbb{A}^2$ and let $U_{i,j} = \{x \in U_i : x_j \neq 0\}$, where a point in X_0 is specified by coordinates (x_1, x_2) , a point in X_1 by (x_0, x_2) , and a point in X_2 by (x_0, x_1) . The space formed by gluing along

$$f_{01}(x_1, x_2) = \left(\frac{1}{x_1}, \frac{x_2}{x_1} \right), f_{02}(x_1, x_2) = \left(\frac{1}{x_2}, \frac{x_1}{x_2} \right), f_{12} = \left(\frac{x_0}{x_1}, \frac{1}{x_1} \right)$$

is the **projective plane** $\mathbb{P}^2$.

Since every prevariety X is finitely covered by affine varieties, we can take an open cover U_i and set $U_{ij} = U_i \cap U_j$ open, with $f_{ij} : U_i \rightarrow U_j$ given by the identity. This recovers X , so

this construction gives any prevariety in terms of affine varieties.

Since a prevariety is a ringed space, in particular it is topological. Moreover, it is the finite union of Noetherian spaces, so by homework it is Noetherian. Moreover, we can endow any open or closed subset with a prevariety structure.

Definition 2.16

Let $U \subseteq X$ be an open subset of a prevariety. Then U is a ringed space endowed with the structure sheaf

$$\mathcal{O}_U(V) = \mathcal{O}_X(V)$$

Moreover, If X is covered by finitely many affine varieties $X_1, \dots, X_n$, then $U \cap X_i$ is open in X_i , hence covered by finitely many distinguished open subsets, which are affine varieties. So U is also covered by finitely many affine varieties. Endowed with this structure, U is called an **open subprevariety** of X .

Definition 2.17

Let $Y \subseteq X$ be a closed subset of a prevariety. Then Y is a ringed space endowed with the structure sheaf $\mathcal{O}_Y(V)$ of functions which are locally restrictions of regular functions:

$$\{\varphi : V \rightarrow K : \forall a \in V, \exists U_a \subseteq X \text{ open } \exists \psi \in \mathcal{O}_X(U_a) \text{ s.t. } \varphi = \psi \text{ on } U_a \cap V\}$$

This is called a **closed subprevariety**.

Proposition 2.25

For any affine open set $W \subseteq X$ and closed set $Y \subseteq X$, the identity map on $W \cap Y$ is an isomorphism of ringed spaces between $W \cap Y$ equipped with the sheaf $\mathcal{O}_Y|_W$ (from the closed subprevariety construction) and the equipped with the sheaf $\mathcal{O}_{W \cap Y}$ (treating $W \cap Y$ as an affine subvariety of W).

Proof. The identity is clearly continuous and its own inverse. So we just need to show it is a morphism in both directions. $\square$

Corollary 2.26

If $Y \subseteq X$ is a closed subset of a prevariety, then $(Y, \mathcal{O}_Y)$ as defined above is a prevariety.

Proof. Cover X by finitely many affine varieties X_i . Then $(Y, \mathcal{O}_Y)$ is covered by $Y \cap X_i$ with the sheaf $\mathcal{O}_Y|_{Y \cap X_i}$. By the proposition, this is isomorphic to $(Y \cap X_i, \mathcal{O}_{Y \cap X_i})$ which is an affine variety. So $(Y, \mathcal{O}_Y)$ is covered by finitely many affine varieties. $\square$

The finite union of open or closed subprevarieties is also an open or closed subprevariety, respectively. However, unioning an open and closed subprevariety together does not in general give a prevariety.

Example 2.15

Let $X = \mathbb{A}^1 \times (\mathbb{A}^1 \setminus \{0\})$, $Y = \{(0, 0)\} \subseteq \mathbb{A}^2$, and let $Z = X \cup Y$. X is open and Y closed in $\mathbb{A}^2$, so they are subprevarieties. However, there is no neighborhood of the origin that looks like an affine variety.

To see this, let U be an open neighborhood of the origin in Z , which is therefore of the form $Z \cap V$ for V open in $\mathbb{A}^2$. Then $V \cap D(y)$ is a nonempty open set, hence dense in $\mathbb{A}^2$, and therefore it is also irreducible by irreducibility of $\mathbb{A}^2$. $V \cap D(y)$ is also contained in U so its closure in U is U , thus it is dense in X . Therefore U is also irreducible.

The dimension of U is two, since it lies in $\mathbb{A}^2$ and we have the strict chain

$$\{(0, 0)\} \subsetneq U \cap V(x) \subsetneq U$$

(note that $U \cap V(x)$ is nonempty and open in $V(x) \cong \mathbb{A}^1$, which is irreducible, hence irreducible). Then y is in the coordinate ring of U , so its vanishing locus has codimension 1, implying it has dimension 1. But it is a single point so has dimension 0, contradiction.

Proposition 2.27

Let $Y \subseteq X$ be a closed subprevariety. Then

1. The inclusion $i_Y : Y \rightarrow X$ is a morphism,
2. If $f : Z \rightarrow X$ is a morphism and $f(Z) \subseteq Y$ where Z is a prevariety, then the same set theoretic map is a morphism from $Z \rightarrow Y$.

Proof. 1. The inclusion is closed by definition of the subspace topology. It is a morphism since if $\varphi \in \mathcal{O}_X(U)$, then $i_Y^* \varphi$ is a map from $i_Y^{-1}(U) \rightarrow K$. Since i_Y is the restriction of the identity to Y , for any $a \in i_Y^{-1}(U)$ we have $a \in U \subseteq X$ with $\varphi = \varphi$ on $U \cap i_Y^{-1}(U)$. So i_Y is a morphism.

2. f is clearly still continuous, and if $\varphi \in \mathcal{O}_Y(U)$, then at any point $a \in U$ we have $a \in U_a \subseteq X$ as well as $\psi \in \mathcal{O}_X(U_a)$ with $\varphi = \psi$ on $U \cap U_a$. $f^* \psi \in \mathcal{O}_Z(f^{-1}(U_a))$ and since $f(Z) \subseteq Y$, $f^{-1}(U_a) = f^{-1}(U \cap U_a)$. Then we have $f^* \psi = f^* \varphi \in \mathcal{O}_Z(f^{-1}(U \cap U_a))$. Taking the union over all a , $f^* \varphi$ is regular on an open cover so it is regular. Thus $f^* \varphi \in \mathcal{O}_Z(f^{-1}(U))$, meaning that f is a morphism. $\square$

Proposition 2.28

The inverse image of an open (resp. closed) subprevariety under a morphism is an open (closed) subprevariety.

Proof. Straightforward by continuity. $\square$

Example 2.16

Let $f : \mathbb{A}^2 \rightarrow \mathbb{A}^2$ be $(x, y) \mapsto (xy, y)$. Then $f(\mathbb{A}^2)$ is the set $Z = \mathbb{A}^1 \times (\mathbb{A}^1 \setminus \{0\}) \cup \{(0, 0)\}$ that we investigated earlier, which is neither an open nor closed subprevariety. So this property is not preserved under forward images.

The union of a closed and open subset is a **locally closed** subset, and a **constructible set** is a finite union of locally closed sets. We can show that the forward image of a constructible set under a morphism is constructible.

Finally, we study products of prevarieties, because this will allow us to state the condition that ensures a variety avoids the pathological behavior of some prevarieties. If X, Y are prevarieties, then we want to make sense of the product $X \times Y$. Set theoretically this is no problem, but we need to endow it with a ringed space structure.

The only way that we can currently build up prevarieties is by gluing them together. One way that we can do this is to pick affine covers U_i of X and V_j of Y , and glue together the sets $U_i \cap V_j$, which we do know have prevariety structures as affine varieties. However, this is not a priori independent of the chosen covers. We will solve this by defining the product via a universal property, so it is unique up to unique isomorphism.

Definition 2.18

Let X, Y be prevarieties. A **product of prevarieties** is a prevariety P with morphisms $\pi_X : P \rightarrow X, \pi_Y : P \rightarrow Y$ such that for any prevariety Z with morphisms $f_X : Z \rightarrow X, f_Y : Z \rightarrow Y$ there is a unique morphism $f : Z \rightarrow P$ such that the following commutes:

$$\begin{array}{ccccc} Z & & & & \\ & \searrow \exists! f & & \searrow f_Y & \\ & & P & \xrightarrow{\pi_Y} & Y \\ & \swarrow f_X & \downarrow \pi_X & & \\ & & X & & \end{array}$$

In particular, when $P = X \times Y$ and Z is assumed to be affine, we showed that this holds.

Proposition 2.29

If X, Y are prevarieties then there is a product P of X and Y , which is unique up to unique isomorphism.

Proof. First we show uniqueness. Let P be a product and P' another product. Applying the universal property of P to the projections of P' , and vice versa, we get morphisms

$f : P \rightarrow P', g : P' \rightarrow P$ such that

$$\begin{array}{ccc} P & \xrightarrow{\pi_Y} & Y \\ \pi_X \downarrow & \swarrow g & \uparrow \pi'_Y \\ X & \xleftarrow{\pi'_X} & P' \end{array}$$

f (dashed arrow from P to P')

This implies the following diagram:

$$\begin{array}{ccc} P & \xrightarrow{\pi_Y} & Y \\ \pi_X \downarrow & \searrow g \circ f & \downarrow \pi_X \\ X & & P \end{array}$$

id_P (dashed arrow from P to P)

By the universal property for P , $id_P = g \circ f$, and by the same for P' , $f \circ g = id_{P'}$. So P, P' are isomorphic by a unique isomorphism.

Now, for existence, we let P be constructed as above, where it is glued across affine varieties $U_i \times V_j$. To check that this satisfies the universal property, we have that each $U_i \times V_j$ offers a projection morphism $U_i \times V_j \rightarrow U_i \subseteq X$ and one to $V_j \subseteq Y$. They agree on intersections, so they glue into a morphism $\pi_X : P \rightarrow X, \pi_Y : P \rightarrow Y$. Now, let Z be a prevariety and $f_X : Z \rightarrow X, f_Y : Z \rightarrow Y$ be morphisms. We want to show that

$$f = (f_X, f_Y) : Z \rightarrow P$$

is a morphism. It is clearly unique since this is the only possible way the diagram can commute. It suffices to check this on an open cover. Consider the open sets $f^{-1}(U_i \times V_j)$. These are open subprevarieties, so we can cover them by affine open subsets Z_k . On each Z_k , $f_X|_{Z_k} : Z_k \rightarrow U_i$ is a morphism as the composition of the open embedding morphism with a morphism, and similarly for $f_Y|_{Z_k}$. Then since we are consider maps from affine varieties, the universal property for products of affine varieties applies, so $f|_{Z_k} = (f_X|_{Z_k}, f_Y|_{Z_k})$ is a morphism from $Z_k \rightarrow U_i \times V_j$. By the gluing construction of prevarieties, the embedding $U_i \times V_j \rightarrow P$ is a morphism. We can then glue these into a morphism $f : Z \rightarrow P$. So this is a morphism and the universal property holds. $\square$

Proposition 2.30

If $X' \subseteq X, Y' \subseteq Y$ are closed subprevarieties then $X' \times Y'$ with the product structure is isomorphic to $X' \times Y'$ with the closed prevariety structure inherited from $X \times Y$ with the product structure.

Finally, we conclude this section by defining a variety to be a prevariety which is sufficiently nice to avoid pathological topological spaces like the affine line with two origins. In this case, the set equipped with the complex topology is not Hausdorff. We cannot directly require that varieties are Hausdorff prevarieties, as this is too strong of a requirement (we

cannot find disjoint open neighborhoods). However, it is a result that a topological space X is Hausdorff if and only if its diagonal is closed in the product topology of $X \times X$. Since our products do not carry the product topology, the following is a useful definition even though being Hausdorff is not:

Definition 2.19

A prevariety X is called a **variety** if its diagonal

$$\Delta_X = \{(x, x) : x \in X\} \subseteq X \times X$$

is closed under the Zariski topology.

Lemma 2.31

Affine varieties are varieties.

Proof. If $X \subseteq \mathbb{A}^n$ then we can write the diagonal as the relative vanishing locus

$$\Delta_X = V_{X \times X}(x_1 - y_1, x_2 - y_2, \dots, x_n - y_n) \subseteq X \times X$$

so Δ_X is closed. □

Proposition 2.32

Let $f_1 : Y_1 \rightarrow X_1, f_2 : Y_2 \rightarrow X_2$ be morphisms (of prevarieties). Then $(f_1, f_2) : Y_1 \times Y_2 \rightarrow X_1 \times X_2$ is a morphism.

Proof. The maps $\pi_1 : Y_1 \times Y_2 \rightarrow Y_1, \pi_2 : Y_1 \times Y_2 \rightarrow Y_2$ are morphisms. Let $g_1 = f_1 \circ \pi_1, g_2 = f_2 \circ \pi_2$ are morphisms. Then there is a unique morphism $f = (F_1, F_2)$ such that $F_1 = g_1, F_2 = g_2$ by the universal property of products, which implies that (f_1, f_2) is a morphism. □

Lemma 2.33

Open and closed subprevarieties of varieties are also varieties.

Proof. Consider the inclusion map $i_{Y \times Y} : Y \times Y \rightarrow X \times X$. If we can show that this is a morphism, then it will follow by continuity that the diagonal is closed, since $\Delta_Y = i_{Y \times Y}^{-1}(\Delta_X)$.

To show this, we note that the open/closed embedding $i_Y : Y \rightarrow X$ is a morphism. Then by the proposition above, $i_{Y \times Y} = (i_Y, i_Y)$ (where the inclusions work on each coordinate separately) is a morphism. □

Definition 2.20

A variety of a pure dimension 1 is called a **curve** and variety of pure dimension 2 is a **surface**. If $Y \subseteq X$ is a subvariety, X is pure dimensional, and Y has pure dimension $\dim X - 1$, then Y is a **hypersurface** in X .

Lemma 2.34

Let $f, g : X \rightarrow Y$ be morphisms of prevarieties, with Y a variety. Then:

1. the graph

$$\Gamma_f = \{(x, f(x)) : x \in X\} \subseteq X \times Y$$

is closed in $X \times Y$,

2. the set

$$\{x \in X : f(x) = g(x)\}$$

is closed in X .

Proof. 1. The graph is the preimage of Δ_Y closed under the map

$$(\text{id}, f)$$

which is a morphism by the proposition above.

2. This is the preimage of Δ_Y under the map

$$(f, g)$$

which is a morphism by the proposition above. □

Proposition 2.35

If X, Y are varieties then $X \times Y$ is a variety.

Proof. The diagonal is given by

$$\Delta_{X \times Y} = \{(x, y), (x, y) : x \in X, y \in Y\} \subseteq X \times Y \times X \times Y$$

This is isomorphic to

$$\{(x, x), (y, y)\} \subseteq X \times X \times Y \times Y$$

This is the set $(\Delta_X \times Y \times Y) \cap (X \times X \times \Delta_Y)$ which are both closed, so this is closed. □

Chapter 3

Projective Varieties

3.1 Construction and Homogeneous Polynomials

In the last chapter we introduced the projective line $\mathbb{P}^1$, which we saw was constructed from $\mathbb{A}^1$ by adding “points at infinity.” If we do this for each $\mathbb{A}^n$, it turns out that a large class of varieties lie in some $\mathbb{P}^n$, and that such varieties have beneficial properties. We begin by describing the construction and topology of projective spaces and varieties. We can do this using a gluing procedure as before, but it is more instructive to build this up by hand, transfer results from affine space, and then finally reconnect this with the theory of varieties by imposing a ringed space structure.

Definition 3.1

Projective space of dimension n is defined as a set by the space of all 1-dimensional linear subspaces of K^{n+1} .

Note that every one dimensional linear subspace can be specified by giving a nonzero vector which generates it:

$$\ell = \langle (x_0, \dots, x_n) \rangle$$

Moreover, two vectors which generate the same subspace agree up to a homogeneous rescaling of their coordinates by $\lambda \neq 0$. Thus we can see $\mathbb{P}^n$ as

$$\mathbb{P}^n = K^{n+1} \setminus \{0\} / \sim$$

where $\sim$ is the equivalence relation of agreeing up to homogeneous rescaling by $\lambda \neq 0$.

Definition 3.2

The **homogeneous coordinates** for an equivalence class $[(x_0, \dots, x_n)] \in \mathbb{P}^n$ for $(x_0, \dots, x_n) \in K^{n+1} \setminus \{0\}$ are

$$(x_0 : \dots : x_n)$$

Homogeneous coordinates are defined up to a constant scaling by $\lambda \neq 0$ and are never all zero. We need to connect affine space with projective space, which proceeds as follows:

Consider the map $f : \mathbb{A}^n \rightarrow \mathbb{P}^n$ which takes $(x_1, \dots, x_n) \mapsto (1 : x_0 : \dots : x_n)$. This is injective since the first coordinate prevents scaling. The image is the set

$$U_0 = \text{im } f = \{(1 : x_1 : \dots : x_n)\} = \{(x_0 : x_1 : \dots : x_n) : x_0 \neq 0\}$$

There is also an inverse

$$f^{-1} : U_0 \rightarrow \mathbb{A}^n, \quad (x_0 : x_1 : \dots : x_n) = \left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0} \right)$$

Thus, as a set, we can think of $\mathbb{A}^n$ as a subset of $\mathbb{P}^n$. This is called the **affine part**. Meanwhile, the complement is the set of points in $\mathbb{P}^n$ where $x_0 = 0$, and the other coordinates are also not all zero, defined up to scaling. So

$$\mathbb{P}^n \setminus \mathbb{A}^n = \mathbb{P}^{n-1}$$

where we think of $\mathbb{P}^{n-1}$ as the space of points at infinity in $\mathbb{P}^n$.

Example 3.1

Let $K = \mathbb{C}$. The affine coordinate to homogeneous coordinate map $\pi : \mathbb{C}^{n+1} \setminus \{0\} \rightarrow \mathbb{P}_{\mathbb{C}}^n$ is surjective. If we restrict to $S^n = \{(x_0, \dots, x_n) : |x_0|^2 + \dots + |x_n|^2 = 1\}$, then S is compact and $\pi|_S$ is surjective, so $\mathbb{P}_{\mathbb{C}}^n$ is compact.

We would like to make sense of solutions to algebraic conditions in $\mathbb{P}^n$. Observe that since homogeneous coordinates are only defined up to scaling, polynomials in their coordinates may not be well defined: if $f(x, y) = x^2 - y$ then $(1 : 1) = (2 : 2)$ but

$$f(1, 1) = 0, f(2, 2) = 2$$

so we cannot necessarily detect the vanishing locus directly. However, if all terms are of the same degree, then they will scale together. While the value is still not well defined, we can detect when the function is exactly zero.

Definition 3.3

A polynomial $f \in K[x_0, \dots, x_n]$ is called **homogeneous** of degree d if all its monomials are of all degree d .

As above, we will prefer to formulate this for non-polynomial rings. The graded ring structure generalizes the notion of degree.

Definition 3.4

A **graded ring** is a ring R with abelian additive subgroups $R_1, R_2, \dots$ such that

$$R = \bigoplus_{d \in \mathbb{N}} R_d$$

so that each element of R is a finite sum of elements of R_d , with a unique decomposition

$$f = \sum_{d \in \mathbb{N}} f_d, \quad \text{finitely many nonzero}$$

and whenever $f \in R_d, g \in R_e, fg \in R_{d+e}$. For $f \neq 0 \in R$, the **degree** of f is

$$\deg(f) = \max \{d : f_d \neq 0\}$$

The nonzero elements of each R_d are called **homogeneous elements**. A graded ring where $\lambda f \in R_d$ whenever $\lambda \in K, f \in R_d$, and R is a K -algebra is called a **graded K -algebra**.

Clearly the polynomials, where R_d denote those polynomials where all monomials are of degree d , form a graded ring.

Proposition 3.1

If $f, g \in R$ are nonzero in a graded ring then

$$\deg(fg) = \deg f + \deg g$$

Proposition 3.2

If R is a nonzero graded ring, then 1 is homogeneous of degree 0.

Proof. There is some R_d which is nonzero, so let $x \in R_d$ be nonzero. Then the homogeneous decomposition of $x \cdot 1$ is just

$$x \cdot 1 = x \in R_d$$

so $(x \cdot 1)_d = x$ and zero for every other component. Thus $\deg(x \cdot 1) = \deg x$ so $\deg 1 = 0$. $\square$

Corollary 3.3

The graded ring structure on $K[x_0, \dots, x_n]$ is uniquely determined by the values $\deg(x_0), \dots, \deg(x_n)$.

Definition 3.5

An ideal of a graded ring is called a **homogeneous ideal** if it can be generated by homogeneous elements (not necessarily of the same degree).

Lemma 3.4

Let J, J_1, J_2 be ideals in a graded ring R .

(a) J is homogeneous if and only if

$$f = \sum_d f_d \in J \implies f_d \in J \forall d$$

or equivalently if $J = \bigoplus_d (J \cap R_d)$.

(b) If J_1, J_2 are homogeneous, then $J_1 + J_2, J_1 J_2, J_1 \cap J_2, \sqrt{J_1}$ are homogeneous.

(c) If J is homogeneous then R/J is a graded ring with decomposition

$$R/J = \bigoplus_d R_d/J \cap R_d$$

Proof. We first observe that we have the convolution

$$(fg)_d = \sum_{d_1+d_2=d} f_{d_1} g_{d_2}$$

(a) ($\implies$) Assume J is homogeneous generated by the elements h^i . Then $f \in J$ is a finite sum

$$f = \sum_{i \in I} g^i h^i$$

where g^i are not necessarily homogeneous. But we have that the homogeneous decomposition of f is

$$f_d = \sum_{i \in I} g_{d-\deg(h^i)}^i h^i$$

and only finitely many are nonzero, so $f_d \in J$.

($\impliedby$) Assume R is a graded ring such that $f \in J \implies f_d \in J$. Then we want to say that $J = \langle f_d : f \in J, d \in \mathbb{N} \rangle$ is generated by all homogeneous elements. Indeed, each f_d is contained in J by assumption, and each f can be written as a finite sum of its homogeneous elements. So J is homogeneous.

(b) For the first two, let $J_1 = \langle \{h^{(i)} : i \in I_1\} \rangle, J_2 = \langle \{g^{(j)} : j \in I_2\} \rangle$. Then

$$\begin{aligned} J_1 + J_2 &= \left\langle \left\{ h^{(i)} : i \in I_1 \right\} \cup \left\{ g^{(j)} : j \in I_2 \right\} \right\rangle \\ J_1 J_2 &= \left\langle \left\{ h^{(i)} g^{(j)} : i \in I_1, j \in I_2 \right\} \right\rangle \end{aligned}$$

which are all homogeneous. For $J_1 \cap J_2$, we have that

$$f \in J_1 \cap J_2 \implies \begin{cases} f \in J_1 \\ f \in J_2 \end{cases} \implies \forall d, \begin{cases} f_d \in J_1 \\ f_d \in J_2 \end{cases} \implies \forall d, f_d \in J_1 \cap J_2$$

so $J_1 \cap J_2$ is homogeneous by part (a). For $\sqrt{J}$, we use the condition from (a). If $f \in \sqrt{J}$, write

$$f = f_0 + \dots + f_d$$

and induct on d . If $f^m \in J$, then

$$(f^m)_{md} = f_d^m$$

which is in J by part (a). So $f_d \in \sqrt{J}$, meaning $f - f_d \in \sqrt{J}$ as well. Inducting, we find that all f_d lie in $\sqrt{J}$. So $\sqrt{J}$ is homogeneous.

- (c) We need to show that $R_d \cap J$ is a subgroup of R_d , and that there exists a unique homogeneous decomposition in R/J . For the first, we restrict the projection $R \rightarrow R/J$ to R_d to get $\varphi : R_d \rightarrow R/J$ a group homomorphism with kernel $R_d \cap J$. Then

$$R_d / R_d \cap J \cong \text{im}(\varphi_d) \subseteq R_d$$

which is a subgroup. The existence of a homogeneous decomposition is clear since if

$$f = \sum_d f_d \in R$$

then

$$\bar{f} = \sum_d \bar{f}_d \in R/J$$

and uniqueness follows since If

$$\sum_d \bar{f}_d = \sum_d \bar{g}_d \in R/J$$

then

$$\Delta = \sum_d f_d - g_d \in J$$

By part (a), we then have

$$\Delta_d = f_d - g_d \in J \implies \bar{f}_d = \bar{g}_d \in R_d / R_d \cap J \quad \square$$

We will use homogeneous ideals to define formulate projective subvarieties and a projective version of the Nullstellensatz.

Definition 3.6

Let $S \subseteq K[x_0, \dots, x_n]$ be a set of homogeneous polynomials. Then the (projective) **vanishing locus** of S is

$$V_p(S) = V(S) = \{x \in \mathbb{P}^n : f(x) = 0 \forall f \in S\}$$

Sets of the form $V(S)$ are called **projective varieties**. If J is a homogeneous ideal in $K[x_0, \dots, x_n]$ then

$$V_p(J) = V(J) = \{x \in \mathbb{P}^n : f(x) = 0 \forall f \in J \text{ homogeneous}\}$$

Equivalently, if S is the set of homogeneous polynomials in J , then $J = \langle S \rangle$ and $V(J) = V(S)$. If $X \subseteq \mathbb{P}^n$ then we define the ideal of X to be

$$I_p(X) = I(X) = \langle f \in K[x_0, \dots, x_n] : f(x) = 0 \forall x \in X \rangle$$

When we mix the definitions, the affine vanishing locus and ideal are denoted V_a, I_a .

Example 3.2

$\emptyset = V_p(1)$ and $\mathbb{P}^n = V_p(0)$ are both projective varieties.

Definition 3.7

If $f_1, \dots, f_r$ are homogeneous linear polynomials then $V_p(f_1, \dots, f_r)$ is called a **linear subspace** of $\mathbb{P}^n$.

Proposition 3.5

A point $\{a\} \in \mathbb{P}^n$ is a projective variety.

Proof.

□

Example 3.3

Consider $f = x_1^2 - x_2^2 - x_0^2$. In $\mathbb{A}^3$, the real part of $V_a(f)$ looks like a cone. If we intersect with $\{x_0 = 1\}$ and consider which linear subspaces pass through $V_a(f) \cap \{x_0 = 1\}$, the real part looks like a hyperbola. In projective space, we add two more linear subspaces which are those contained in $V_a(f) \cap \{x_0 = 0\}$. These connect the hyperbola so that the real part of $V_p(f)$ is a circle.

This shows that V_a, V_p behave similarly, and many of the results for $V_a(f)$ should be able to be carried over $V_p(f)$.

Definition 3.8

An affine variety $X \subseteq \mathbb{A}^{n+1}$ is called a **cone** if $0 \in X$ and $\lambda x \in X$ whenever $\lambda \in K, x \in X$. Equivalently, the first condition can be taken to be $X \neq \emptyset$. If X is a cone then the **projectivization** of X is the set

$$\mathbb{P}(X) = \pi(X \setminus \{0\}) \subseteq \mathbb{P}^n$$

where $\pi : \mathbb{A}^{n+1} \rightarrow \mathbb{P}^n$ is defined by

$$(x_0, \dots, x_n) \mapsto (x_0 : \dots : x_n)$$

If $X \subseteq \mathbb{P}^n$ is a projective variety then the cone over X is

$$C(X) = \pi^{-1}(X) \cup \{0\}$$

Proposition 3.6

Let $S \subseteq K[x_0, \dots, x_n]$ be a set of homogeneous nonconstant polynomials. Then $V_a(S)$ is a cone.

Proof. $0 \in V_a(S)$ since every homogeneous nonconstant polynomial vanishes at zero. If $x \in V_a(S)$ then for any $f \in S$ of degree d ,

$$f(\lambda x) = \lambda^d f(x) = 0$$

so $\lambda x \in V_a(S)$. □

Proposition 3.7

If $X \subseteq \mathbb{A}^{n+1}$ is a cone then $I(X) = I_a(X)$ is a homogeneous ideal in $K[x_0, \dots, x_n]$.

Proof. From before, we need to show that when $f \in I(X), f_d \in I(X)$. For any $x \in X$, we have

$$0 = f(\lambda x) = \sum_{d \in \mathbb{N}} \lambda^d f_d(x)$$

This is a polynomial in λ which vanishes for all λ , so using the fact that K is algebraically closed, $f_d(x) = 0$ for all d . This follows for all x , so $f_d \in I(X)$ for all d , so $I(X)$ is homogeneous. □

Lemma 3.8

If S is a set of homogeneous nonconstant polynomials then

$$\mathbb{P}(V_a(S)) = V_p(S), \quad C(V_p(S)) = V_a(S)$$

Theorem 3.9

The maps $\mathbb{P}, C$ define a bijection between cones in $\mathbb{A}^{n+1}$ and projective varieties in $\mathbb{P}^n$.

Proof. First we show that this is a well defined bijection. If $X \subseteq \mathbb{A}^{n+1}$ is a cone then by the Nullstellensatz,

$$X = V_a(I_a(X))$$

Define $J = I_a(X)$ and let S be its homogeneous elements. Since J is homogeneous, we have

$$X = V_a(J) = V_a(S)$$

Then the projectivization satisfies

$$\mathbb{P}(X) = \mathbb{P}(V_a(J)) = V_p(J)$$

In the other direction, if Y is a projective variety then let $Y = V_p(J)$ with homogeneous elements S . If J contains constant polynomials then $J = K[x_0, \dots, x_n]$ and $Y = \emptyset$, and $C(Y) = \{0\}$ is a cone. Otherwise, we showed $V_a(S)$ is a cone, and

$$C(Y) = C(V_p(J)) = C(V_p(S)) = V_a(S)$$

To show the bijection, let X be a cone. X is nonempty so $I_a(X)$ does not contain the constants. Then

$$C(\mathbb{P}(X)) = C(\mathbb{P}(V_a(I_a(X)))) = C(V_p(I_a(X))) = V_a(I_a(X)) = X$$

and for Y a projective variety

$$\mathbb{P}(C(Y)) = \pi(\pi^{-1}(Y)) = Y \quad \square$$

Now we would like to formulate a projective Nullstellensatz. The naive statement is unfortunately not true, as the edge cases in the previous results have shown: if $X = V_p(K[x_0, \dots, x_n]) = \emptyset$, then

$$C(X) = \{0\} \implies I_a(C(X)) = \langle x_0, \dots, x_n \rangle$$

which shows that the ideal $I_0 = \langle x_0, \dots, x_n \rangle$ is somehow related to our edge cases from before. Indeed, $V_p(\langle x_0, \dots, x_n \rangle) = \pi(V_a(\langle x_0, \dots, x_n \rangle) \setminus \{0\}) = \pi(\{0\} \setminus \{0\}) = \emptyset$, so

$$I_p(V_p(\langle x_0, \dots, x_n \rangle)) = I_p(\emptyset) = K[x_0, \dots, x_n] \neq \sqrt{\langle x_0, \dots, x_n \rangle}$$

Definition 3.9

The **irrelevant ideal** is the ideal $I_0 = \langle x_0, \dots, x_n \rangle$.

Theorem 3.10: Projective Nullstellensatz

- (a) If X is a projective variety then $V_p(I_p(X))$.
- (b) If $J \subseteq K[x_0, \dots, x_n]$ is a homogeneous ideal and $\sqrt{J}$ is not the irrelevant ideal then $I_p(V_p(J)) = \sqrt{J}$.

This defines an inclusion reversing bijection between projective varieties and radical homogeneous ideals besides the irrelevant ideal.

Proof. (a), and the inclusion reversing part of the statement are the same as in the affine case. Also, the proof of $I_p(V_p(J)) \supseteq \sqrt{J}$ is the same as the affine case.

For the other inclusion, assume J is homogeneous and $\sqrt{J} \neq I_0$. If J contains any constant polynomials then $J = K[x_0, \dots, x_n]$, and then we have

$$I_p(V_p(K[x_0, \dots, x_n])) = I_p(\emptyset) = K[x_0, \dots, x_n] = \sqrt{J}$$

Otherwise, write $X = V_a(J) \subseteq \mathbb{A}^{n+1}$, $Y = V_p(J) \subseteq \mathbb{P}^n$. Since J does not contain the constants, we know from before that

$$\mathbb{P}(X) = Y, C(Y) = X$$

By the affine Nullstellensatz, we have

$$I_a(X) = \sqrt{J} \neq I_0 = I_a(\{0\})$$

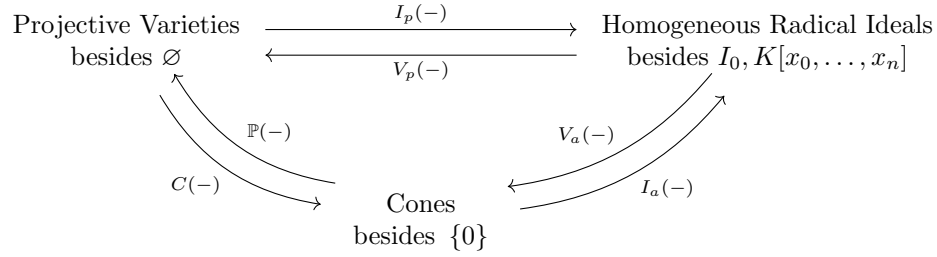
so $X \neq \{0\}$, which implies that $Y \neq \emptyset$ (since we know cones biject with projective varieties, and the cone $\{0\}$ bijects with $\emptyset$). As a result, every constant polynomial in $I_p(Y)$ is the zero polynomial. Moreover, the homogeneous polynomials vanishing on Y (which generate $I_p(Y)$) are all either degree > 0 (in which case they vanish at zero), or the zero polynomial. So such polynomials vanish not only on $X \setminus \{0\}$ (since X is a cone), but also 0 itself, and thus all of X . Thus we have the equalities

$$\begin{aligned} I_p(Y) &= \langle \{f \in K[x_0, \dots, x_n] \text{ homog.} : f|_Y \equiv 0\} \rangle \\ &= \langle \{f \in K[x_0, \dots, x_n] \text{ homog.} : f|_{X \setminus \{0\}} \equiv 0\} \rangle \\ &= \langle \{f \in K[x_0, \dots, x_n] \text{ homog.} : f|_X \equiv 0\} \rangle \end{aligned}$$

X is a cone, so the homogeneous polynomials generate $I_a(X)$, and we conclude that $I_p(Y) = I_a(X) = \sqrt{J}$ by the affine Nullstellensatz. $\square$

As a result, we basically have shown a bijection between projective varieties, cones, and homogeneous radical ideals. However, in each case, we had to exclude some exceptional cases: the cone $\{0\}$, the empty set in $\mathbb{P}^n$, and the ideals $I_0, K[x_0, \dots, x_n]$. This is summarized

in the following diagram:



Proposition 3.11

(a) If S_1, S_2 are sets of homogeneous polynomials in $K[x_0, \dots, x_n]$ then

$$V_p(S_1) \cup V_p(S_2) = V_p(S_1 S_2)$$

and for any collection (S_i) of such sets,

$$\bigcap_{i \in I} V_p(S_i) = V_p\left(\bigcup_{i \in I} S_i\right)$$

(b) If J_1, J_2 are homogeneous ideals of $K[x_0, \dots, x_n]$ then

$$\begin{aligned} V_p(J_1) \cup V_p(J_2) &= V_p(J_1 J_2) = V_p(J_1 \cap J_2) \\ V_p(J_1) \cap V_p(J_2) &= V_p(J_1 + J_2) \end{aligned}$$

(c) If $X_1, X_2 \subseteq \mathbb{P}^n$ are projective varieties then

$$\begin{aligned} I_p(X_1 \cup X_2) &= I_p(X_1) \cap I_p(X_2) \\ I_p(X_1 \cap X_2) &= \begin{cases} \sqrt{I_p(X_1) + I_p(X_2)}, & X_1 \cap X_2 \neq \emptyset \\ K[x_0, \dots, x_n], & X_1 \cap X_2 = \emptyset \end{cases} \end{aligned}$$

3.2 Projective Subvarieties

We now build the coordinate ring of a variety and consider subvarieties.

Definition 3.10

If $Y \subseteq \mathbb{P}^n$ is a variety then its **(projective) coordinate ring** is

$$S(Y) = K[x_0, \dots, x_n] / I_p(Y)$$

By a previous lemma,

$$S(Y) = \bigoplus_{d=0}^{\infty} K_d[x_0, \dots, x_n] / K_d[x_0, \dots, x_n] \cap I_p(Y)$$

is a graded ring, where $K_d[x_0, \dots, x_n]$ denotes the homogeneous polynomials of degree d .

Example 3.4

If Y is the linear subspace $V(x_0 + x_1 + x_2) \subseteq \mathbb{P}^2$, then

$$S(Y) = K[x_0, x_1, x_2] / \langle x_0 + x_1 + x_2 \rangle \cong K[x_0, x_1] = S(\mathbb{P}^1)$$

We will see later that, in analogy with the affine case, this means they are isomorphic as ringed spaces, (hence as projective varieties and varieties).

Recall that, as before, these elements do not make sense as full functions on Y . However, any element of $f \in S(Y)$ with a homogeneous representative does have a well defined vanishing locus on Y . The condition of having a homogeneous representative is equivalent to being homogeneous in the quotient graded ring structure.

Definition 3.11

Let $Y \subseteq \mathbb{P}^n$ be a projective variety. If $J \subseteq S(Y)$ is a homogeneous ideal then its **(relative) vanishing locus** is

$$V_Y(J) = V_{Y,p}(J) = \{x \in Y : f(x) = 0 \forall f \text{ homog.} \in J\}$$

Sets of this form are called **projective subvarieties**. If $X \subseteq Y$ is a subset then its **(relative) vanishing ideal** is

$$I_Y(X) = I_{Y,p}(X) = \langle f \in S(Y) \text{ homog.} : f(x) = 0 \forall x \in X \rangle$$

Example 3.5

For $Y = V(x_0 + x_1 + x_2) \in \mathbb{P}^2$, we have

$$V_Y(-x_0^2 + x_1^2 + x_2^2) = V(x_0 + x_1 + x_2, -(x_1 + x_2)^2 + x_1^2 + x_2^2) = V(x_0 + x_1 + x_2, -2x_1x_2)$$

If K is not of characteristic 2 then we have $x_1 = 0$ or $x_2 = 0$, from which we deduce that

$$V_Y(-x_0^2 + x_1^2 + x_2^2) = \{(1 : 0 : -1), (1 : -1 : 0)\}$$

There is a relative version of the projective Nullstellensatz as well. Also, it can be useful to know that every projective subvariety $X \subseteq Y \subseteq \mathbb{P}^n$ can be written as the vanishing locus

of finitely many homogeneous polynomials in $S(Y)$ of the same degree, since

$$V_p(f) = \emptyset \cup V_p(f) = V_p(\langle x_0^d, \dots, x_n^d \rangle) \cup V_p(f) = V_p(\langle x_0^d f, \dots, x_n^d f \rangle)$$

By the properties of the relative vanishing locus, which are the same as the nonrelative projective versions, we see that finite unions and arbitrary intersections of subvarieties are subvarieties.

Definition 3.12

The **Zariski topology** on Y is the topology where the closed sets are the subvarieties.

This is the same as the relative topology on Y . We want to be able to make a similar statement about $\mathbb{A}^n$, namely that since $\mathbb{A}^n \subseteq \mathbb{P}^n$ as a set, we should also have that the Zariski topology on $\mathbb{A}^n$ is the relative topology as well. Algebraically, this will need to involve some relationship between polynomials in $x_1, \dots, x_n$ and homogeneous polynomials in $x_0, \dots, x_n$.

Definition 3.13

If $f \in K[x_0, \dots, x_n]$ is a homogeneous polynomial then its **dehomogenization** is the (not necessarily homogeneous polynomial) in $K[x_1, \dots, x_n]$ obtained by setting $x_0 = 1$. This is a ring homomorphism as $(f + g)^i = f^i + g^i$, $(fg)^i = f^i g^i$, and since it is surjective, applying this to ideals gives an ideal. Thus we define

$$J^i = \{f^i : f \in J\}$$

for J a homogeneous ideal, which is an ideal in $K[x_1, \dots, x_n]$.

Definition 3.14

If $f \in K[x_1, \dots, x_n]$ is nonzero of degree d , then its **homogenization** is the polynomial

$$f^h = \sum_{k=0}^d x_0^{d-k} f_k$$

where as before, f_k is the degree k homogeneous part.

Since f^h is defined relative to the degree, we have $(fg)^h = f^h g^h$ but $(f + g)^h \neq f^h + g^h$. Therefore to homogenize an ideal, we need to take the generated ideal:

$$J^h = \langle f^h : f \neq 0 \in J \rangle$$

Proposition 3.12

If S is a set of homogeneous polynomials in $K[x_0, \dots, x_n]$ then

$$\langle S \rangle^i = \langle S^i \rangle$$

where $S^i = \{f^i : f \in S\}$.

Proof. Any element $f \in \langle S \rangle$ can be written

$$f = \sum_{n=1}^k s_n$$

where $s_n \in S$. Then $f^i \in \langle S \rangle^i$ and

$$f^i = \sum_{n=1}^k s_n^i \in \langle S^i \rangle$$

In the other direction, if $g \in \langle S^i \rangle$ then

$$g = \sum_{n=1}^k s_n^i \implies g = \left(\sum_{n=1}^k s_n \right)^i \in \langle S \rangle^i$$

□

Example 3.6

It is not true that if $S \subseteq K[x_1, \dots, x_n]$ is a set of nonzero polynomials, then $\langle S \rangle^h = \langle S^h \rangle$. For a counterexample, consider $S = \{x_1\}$, then $S^h = S$ and $\langle S^h \rangle = \langle x_1 \rangle \subseteq K[x_0, \dots, x_n]$. But

$$\langle S \rangle^h = \langle f^h : f \in \langle S \rangle \rangle$$

In particular, $x_1 + x_1x_2 \in \langle S \rangle$, but its homogenization is $x_0x_1 + x_1x_2$ which is not in $\langle x_1 \rangle$.

Homogenization and dehomogenization gives the connection that we need, since it relates the behavior of polynomials under the restriction $x_0 \neq 0$, which restricts $\mathbb{P}^n$ to $\mathbb{A}^n$ under our identification. That is, if F is the map

$$(x_1, \dots, x_n) \rightarrow (1 : x_1 : \dots : x_n)$$

then for homogeneous $f \in K[x_0, \dots, x_n]$,

$$f(F(x_1, \dots, x_n)) = f^i(x_1, \dots, x_n)$$

and for $g \in K[x_1, \dots, x_n]$ nonzero,

$$g^h(F(x_1, \dots, x_n)) = g(x_1, \dots, x_n)$$

Theorem 3.13

The subspace topology on $\mathbb{A}^n \subseteq \mathbb{P}^n$ is the Zariski topology on $\mathbb{A}^n$.

Proof. Observe that if $X = V_p(J) \cap \mathbb{A}^n$ is closed in the subspace topology, then by the remarks above, $X = V_a(J^i)$ is closed in the Zariski topology. On the other hand, if $X = V_a(J)$ is Zariski closed then $X = V_p(J^h) \cap \mathbb{A}^n$ is closed in the subspace topology. $\square$

Now that we have found an embedding of affine space in projective space, we can transfer many of the results we found before.

Proposition 3.14

$\dim \mathbb{P}^n = n.$

Proof. We just showed the subspace

$$U_0 = \{x_0 = 0\}$$

is isomorphic to $\mathbb{A}^n$. This is also true for each $U_i = \{x_i = 0\}$, and collectively the U_i cover $\mathbb{P}^n$. So $\mathbb{P}^n$ is covered open irreducible sets of dimension n , and thus $\dim \mathbb{P}^n = n$. $\square$

Proposition 3.15

If X is a projective variety, then X is irreducible if and only if $S(X)$ is an integral domain.

Proposition 3.16

If X, Y are nonempty projective varieties in $\mathbb{P}^n$, and $\dim X + \dim Y \geq n$, then $X \cap Y \neq \emptyset$.

We have the intuition that an affine variety can be compactified into a projective variety by adding points “at infinity.” This notion is formalized by taking the projective closure of a variety. If an affine variety has $X = V_a(J)$, then we know also that $X = V_p(J^h) \cap \mathbb{A}^n$. This gives $V_p(J^h)$ as a candidate for the closure.

Proposition 3.17

Let $J \subseteq K[x_1, \dots, x_n]$ be an ideal, and let $X = V_a(J)$. Then the closure of X in $\mathbb{P}^n$ (via the embedding $\mathbb{A}^n \hookrightarrow U_0$) is

$$\overline{X} = V_p(J^h)$$

and if $J = \langle f \rangle$ is nonzero and principal then

$$\overline{X} = V_p(f^h)$$

Proof. We know that $\overline{X} \subseteq V_p(J^h)$ since the latter is a closed set containing X . In the other direction, let $Y = V_p(J')$ be a closed set in $\mathbb{P}^n$ containing X . We want to show that

$V_p(J^h) \subseteq J'$, which is equivalent via the Nullstellensatz to saying that $J' \subseteq \sqrt{J^h}$ (assume J' is radical).

J' is homogeneous, so it is generated by its homogeneous elements. Let $g \in J'$ be homogeneous. Note that by dividing out by the highest power of x_0 dividing g , we have $g = x_0^d f^h$ for some $f \in K[x_1, \dots, x_n]$. This is zero on $X \subseteq \mathbb{P}^n$, and since $X \subseteq \mathbb{A}^n$ as well, we have that $(f^h)^i = f$ is zero on $X \subseteq \mathbb{A}^n$. Thus $f \in I_a(X) = I_a(V_a(J)) = \sqrt{J}$. Then $f^m \in J$ for some J , meaning $(f^m)^h \in J^h$. Homogenization commutes with multiplication so $(f^h)^m \in J^h$, meaning $f^h \in \sqrt{J^h}$ and thus $g \in \sqrt{J^h}$.

To show the result when J is principal, we have

$$\begin{aligned} J = \langle fg : g \in K[x_1, \dots, x_n] \rangle &\implies J^h = \langle (fg)^h : g \in K[x_1, \dots, x_n] \rangle \\ &= \langle f^h g^h : g \in K[x_1, \dots, x_n] \rangle = \langle f^h \rangle \end{aligned} \quad \square$$

Finally, we use this result to prove that, as in the affine case, the ideal of a projective hypersurface is principal.

Definition 3.15

A projective variety $X \subseteq \mathbb{P}^n$ is a **projective hypersurface** if it has pure dimension $n - 1$.

Proposition 3.18

If $X \subseteq \mathbb{P}^n$ is a hypersurface then $I_p(X)$ is principal.

Proof. Say X has irreducible decomposition $X = X_0 \cup \dots \cup X_m$. First assume that no component is contained in $V_p(x_0) = \mathbb{P}^n \setminus \mathbb{A}^n$, the points at infinity. Then

$$X \cap \mathbb{A}^n = \bigcup_{n=1}^{\infty} X_n \cap \mathbb{A}^n$$

Each $X_n \cap \mathbb{A}^n$ is a nonempty set open in the relative topology of X_n . Thus

$$\overline{X \cap \mathbb{A}^n} \supseteq \bigcup_{n=1}^{\infty} X_n = X$$

and clearly the other inclusion is true, too, so the projective closure of $X \cap \mathbb{A}^n$ is X . $X \cap \mathbb{A}^n$ is an affine hypersurface so $X \cap \mathbb{A}^n = V_a(f)$ for some nonzero f , and then by the previous proposition,

$$X = V_p(f^h)$$

In the general case, if $X = X' \cup X_n$ and $X_n \subseteq V_p(x_0)$, then by irreducibility and dimension, $X_n = V_p(x_0)$. Then we have $X' = V_p(f)$ for some f , and

$$X = V_p(x_0 f) \quad \square$$

We make a few remarks on this. First, we can assume that f itself is homogeneous by taking its unique factorization. Second, as we remarked above, it is not true in general that $\langle f, g \rangle^h = \langle f^h, g^h \rangle$, or for more generators. However, this is true if we have a Gröbner basis, which is the basis for computer algebra.

3.3 Projective Varieties as Ringed Spaces

We now impose a ringed space structure on projective varieties and want to show that they are varieties. In particular, they should be compatible with affine space. To start, we want to define regular functions. Recall that for affine varieties, regular functions are locally quotients f/g . For projective space, we need to make a restriction, as even a quotient of two homogeneous polynomials is not well defined (it scales the numerator and denominator differently).

Definition 3.16

Let $U \subseteq X$ be open in X a projective variety. Then $\varphi : U \rightarrow K$ is a **regular function** if it can be written locally as

$$\varphi = \frac{g}{f}$$

for $g, f \in S(X)$ homogeneous of the same degree. The set of regular functions on U is denoted $\mathcal{O}_X(U)$.

Since this is a local condition, we therefore have a sheaf, making $(X, \mathcal{O}_X)$ into a ringed space. Actually, $\mathcal{O}_X(U)$ is also a K -algebra, as

$$\frac{f_1}{g_1} + \frac{f_2}{g_2} = \frac{f_1 g_2 + f_2 g_1}{g_1 g_2}$$

and both the numerator and denominator have degree $d_1 d_2$. Now, we will show that this is a prevariety by using the open cover $\mathbb{P}^n = U_0 \cup \dots \cup U_n$.

Proposition 3.19

Let $X \subseteq \mathbb{P}^n$ be a projective variety, and define

$$U_i = \{(x_0 : \dots : x_n) \in X : x_i \neq 0\}$$

Then each U_i is an affine variety, and in particular X is a prevariety.

Proof. By symmetry we can just show $i = 0$. Say $X = V_p(J)$ for J homogeneous. Define $Y = V_a(J^i)$. We claim that $Y \cong U_0$, in particular via the isomorphism $F : Y \rightarrow U_0$

$$(x_1, \dots, x_n) \mapsto (1 : x_1 : \dots : x_n)$$

with inverse

$$(x_0 : x_1 : \cdots : x_n) \mapsto \left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0} \right)$$

These are well defined maps of sets, and they are continuous since

$$\begin{aligned} F^{-1}(V_p(J') \cap U_0) &= V_{a,Y}((J')^i) \\ F(V_a(\tilde{J})) &= V_p(\tilde{J}^h) \cap U_0 \end{aligned}$$

To see that they pull back regular functions, if $\varphi(x_0, \dots, x_n) = g/f$ locally on U_0 , for $f, g \in S(X)_d$, then

$$F^*\varphi = \frac{g^i}{f^i}$$

which is regular on Y . In the other direction, if $\psi(x_1, \dots, x_n) = g/f$ locally on Y , then

$$(F^{-1})^*\psi = \frac{g\left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}\right)x_0^e}{f\left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}\right)x_0^e}$$

where $e = \max(\deg g, \deg f)$. The numerator and denominator are homogeneous polynomials of the same degree. So F, F^{-1} are morphisms, and we conclude. $\square$

Lemma 3.20

Let $X \subseteq \mathbb{P}^n$ be a projective variety, $f_0, \dots, f_m \in S(X)$ be homogeneous of the same degree. Then letting $U = X \setminus V_p(f_0, \dots, f_m)$, the map $f : U \rightarrow \mathbb{P}^m$ given by

$$x \mapsto (f_0(x) : \cdots : f_m(x))$$

is a morphism.

Proof. This is well defined as a map of sets since the homogeneous functions scale together, and they cannot all vanish simultaneously since we are away from their vanishing locus. By the gluing property of morphisms, it suffices to check this on an open cover. Set $V_i = \{x_i \neq 0\} \subseteq \mathbb{P}^m$ and $U_i = f^{-1}(V_i) = \{f_i(x) \neq 0\} \subseteq U$. These are open and cover U . We know each V_i is isomorphic to $\mathbb{A}^n$, so we get the following diagram:

$$\begin{array}{ccc} U_0 & \xrightarrow{f} & V_0 \\ & \searrow & \uparrow \cong \\ & & \mathbb{A}^n \end{array}$$

$(\frac{f_1}{f_0}, \dots, \frac{f_m}{f_0})$

The map from $U_0 \rightarrow \mathbb{A}^n$ is a map from a prevariety to an affine variety with coordinates given by quotients of polynomials. We previously showed that rational coordinate maps between affine varieties were morphisms, the same is true if we only assume the domain is a prevariety. Thus f is a morphism on each U_i , and we conclude that f is a morphism. $\square$

Example 3.7

For any $A \in \mathrm{GL}_{n+1}(K)$, the map $x \mapsto Ax$ is an automorphism of $\mathbb{P}^n$ with inverse $x \mapsto A^{-1}x$, since each coordinate is homogeneous of degree 1.

Moreover it is true that every projective automorphism of $\mathbb{P}^n$ is of this form, and they are unique up to scaling. If $\mathrm{PGL}_{n+1}(K)$ is $\mathrm{GL}_{n+1}(K)$ quotiented by the identification $\lambda I_{n+1} \sim I_{n+1}$ for all $\lambda \in K$, then

$$\mathrm{Aut}(\mathbb{P}^n) \cong \mathrm{PGL}_{n+1}(K)$$

We can describe another class of morphisms which are geometrically seen as projecting from a point. To begin, we project from the affine origin point $a = (1, 0, \dots, 0)$.

Proposition 3.21

The map $f : \mathbb{P}^n \setminus \{a\} \rightarrow \mathbb{P}^{n-1}$ defined by

$$(x_0 : \dots : x_n) \mapsto (x_1 : \dots : x_n)$$

is a morphism.

Proof. The coordinates are polynomials which do not simultaneously vanish. $\square$

We can view this as a projection by fixing $x = (x_0 : \dots : x_n) \in \mathbb{P}^n \setminus \{a\}$. Then the line through a, x is parameterized by

$$\begin{aligned} L_{ax} &= \{sa + tx : (s : t) \in \mathbb{P}^1\} = \{(s + tx_0 : tx_1 : \dots : tx_n) : (s : t) \in \mathbb{P}^1\} \\ &= \{(s : tx_1 : \dots : tx_n) : (s : t) \in \mathbb{P}^1\} \end{aligned}$$

Then the point $(x_1 : \dots : x_n)$, which is the output of this mapping, is the unique intersection point of L_{ax} with the line at infinity $L = \{x_0 = 0\}$. In this way, we can view f as a projection from a point onto a linear subspace. Up to translation and a linear transformation, which are isomorphisms, any point and linear subspace can be moved to a, L , so this describes projection from any point onto any linear subspace not containing it. In general a line through a, b looks like

$$L_{ab} = \{sa + tb : (s : t) \in \mathbb{P}^1\}$$

We note that the map f as above does not extend to a morphism on all of $\mathbb{P}^n$. However, if we restrict to some subvarieties containing a , an extension is sometimes possible.

Example 3.8

Consider the subvariety $X = V_p(x_0x_2 - x_1^2)$, which contains $a = (1 : 0 : 0)$. Define

$f : X \rightarrow \mathbb{P}^1$ by

$$x = (x_0 : x_1 : x_2) \mapsto \begin{cases} (x_1 : x_2), & x \neq a \\ (x_0 : x_1), & x \neq (0 : 0 : 1) \end{cases}$$

We observe that away from $a, (0 : 0 : 1)$, x_0, x_1 are both nonzero on X , so that on the overlap we have

$$x_0 x_2 - x_1^2 = 0 \implies \frac{x_2}{x_1} = \frac{x_1}{x_0}$$

which says that the two homogeneous coordinates are the same. Thus f locally has a rational representation by homogeneous polynomials of the same degree, and thus it is a morphism. Thus the map above has an extension to a . With some more work, we can show that it is actually an isomorphism, so that $X \cong \mathbb{P}^1$.

Proposition 3.22

Let $X \subseteq \mathbb{P}^2$ be any projective conic over a field of $\text{char}(K) \neq 2$. Then $X \cong \mathbb{P}^1$.

Proof. Over such fields, the affine part of any projective conic is isomorphic to either $V_a(x_2 - x_1^2)$ (the standard parabola) or $V(x_1 x_2 - 1)$ (the standard hyperbola). After taking the projective closure, any projective conic is thus isomorphic to either $V_p(x_2 x_0 - x_1^2)$ or $V_p(x_1 x_2 - x_0^2)$, but of course up to change of coordinates these are the same. So every projective conic is the one we just looked at, and we showed that was isomorphic to $\mathbb{P}^1$. $\square$

Now, we want to prove that projective varieties are varieties. We thus need to understand the diagonal of $\mathbb{P}^n \times \mathbb{P}^n$ (since any closed subprevariety of a variety is a subvariety). We will do this by embedding $\mathbb{P}^n \times \mathbb{P}^n$ in a larger space (and will generalize by taking n, m).

Definition 3.17

Let $n, m \in \mathbb{N}$ and $N = (n+1)(m+1) - 1$. Then the **Segre embedding** of $\mathbb{P}^n \times \mathbb{P}^m$ is the embedding $f : \mathbb{P}^n \times \mathbb{P}^m \rightarrow \mathbb{P}^N$ given by

$$((x_i), (y_j)) = (z_{ij} = x_i y_j)_{0 \leq i \leq n, 0 \leq j \leq m}$$

Note that this scales correctly so it is a well defined map.

Proposition 3.23

If f is the Segre embedding, then $X = f(\mathbb{P}^n \times \mathbb{P}^m)$ is the projective variety

$$X = V_p(z_{ij} z_{kl} - z_{il} z_{kj}, 0 \leq i \leq n, 0 \leq j \leq m)$$

and f is isomorphic with its image, so in particular it is a closed embedding.

Proof. Notice that points of X clearly lie in the vanishing locus since for such points,

$$z_{ij}z_{kl} - z_{il}z_{kj} = x_i y_j x_k y_l - x_i y_l x_k y_j = 0$$

Now, we pass to an affine chart. If $z = (z_{ij})$ satisfies the equations, then there is some nonzero coordinate, and without loss of generality we assume $z_{00} \neq 0$. Then z lies in the affine chart $U_{00} = \{z_{00} \neq 0\} \subseteq \mathbb{P}^N$. Then we can assume $z_{00} = 1$. So for any i, j we have

$$z_{ij} = z_{i0}z_{0j}$$

so we note that this is the image of

$$f((z_{i0})_i, (z_{0j})_j)$$

Thus we have described X .

Now, we want to show f is an isomorphism. First, notice that $f^{-1}(U_{00}) = U_0^{\mathbb{P}^n} \times U_0^{\mathbb{P}^m}$, so in these coordinates we have

$$f(x_1, \dots, x_n, y_1, \dots, y_m) = (x_1, \dots, x_n, y_1, \dots, y_m, (x_i y_j)_{i,j})$$

So f is injective since we can read off its coordinates, and its inverse is given by the projection onto the first $n + m$ coordinates. This is true on all patches U_{ij} , and since any two points lie in some patch containing both, we see that f is globally injective. f, f^{-1} locally have polynomial coordinates so they are morphisms and we are done. $\square$

Corollary 3.24

Every projective variety is a variety.

Proof. As before we just need to show that $\mathbb{P}^n$ is separated. The diagonal is given by

$$\Delta_{\mathbb{P}^n} = \{((x_0 : \dots : x_n), (y_0 : \dots : y_n) : x_i y_j - x_j y_i = 0 \forall i, j)\}$$

To see that this is true, note that the algebraic condition is equivalent to all 2×2 minors of the matrix

$$\begin{bmatrix} x_0 & \cdots & x_n \\ y_0 & \cdots & y_n \end{bmatrix}$$

vanishing, which means the rows are linearly dependent. Since they are nonzero, we have that $x = y$ up to scalar multiples, so that it lies in the diagonal. The reverse inclusion is obvious.

In Segre coordinates, this becomes

$$\Delta_{\mathbb{P}^n} = f^{-1}(V_p(z_{ij} - z_{ji}))$$

so this is the preimage of a closed set in $\mathbb{P}^N$, which is therefore closed. $\square$

We remark that this finally tells us that the product of projective varieties in a projective variety. If $X \subseteq \mathbb{P}^n, Y \subseteq \mathbb{P}^m$, then $X \times Y$ lies in $\mathbb{P}^n \times \mathbb{P}^m$ via a closed embedding (the product of two embeddings), and this lies in $\mathbb{P}^N$ via another closed embedding. So $X \times Y$ is isomorphic to a closed subset of $\mathbb{P}^N$ and is a projective variety.

Now, we will develop the notion of completeness. This generalizes the observation that every projective variety over $\mathbb{C}$ is compact in the Euclidean topology. Compactness itself is not useful since every prevariety is Zariski compact. However, we will see that the image under a morphism of a complete set is complete, analogous to compact sets under continuous maps.

Definition 3.18

A map $f : X \rightarrow Y$ between topological spaces is a **closed map** if $f(V) \subseteq Y$ is closed for every $V \subseteq X$ closed.

Theorem 3.25

The projection map $\pi : \mathbb{P}^n \times \mathbb{P}^m \rightarrow \mathbb{P}^m$ is a closed map.

Proof. Fix $Z \subseteq \mathbb{P}^n \times \mathbb{P}^m$ closed. This is also closed under the Segre embedding, so it is $V_p(f_1, \dots, f_r)$ for each f_k homogeneous in z_{ij} (equivalently, bihomogeneous in x_i, y_j). We remarked earlier that we can take these to be homogeneous of the same degree d . Fix $a \in \mathbb{P}^m$ and set $g_i(\cdot) = f_i(\cdot, a)$. Then:

$$\begin{aligned} a \notin \pi(Z) &\iff V_p(g_1, \dots, g_r) = \emptyset \\ &\iff \sqrt{\langle g_1, \dots, g_r \rangle} = K[x_0, \dots, x_n] \text{ or } \langle x_0, \dots, x_n \rangle \end{aligned}$$

In either case, for each i we have $x_i^{k_i} \in \langle g_1, \dots, g_r \rangle$ for some k_i . In particular, for any monomial of degree $k \geq k_0 + \dots + k_n$, some factor x_i must be raised to a power at least k_i . Thus for such k , $K[x_0, \dots, x_n]_k = \langle g_1, \dots, g_r \rangle_k$; that is, the degree k homogeneous part of $\langle g_1, \dots, g_r \rangle$ is all of the degree k homogeneous polynomials. We also observe that $k \geq d$ since each g_i is homogeneous of degree d , $\langle g_1, \dots, g_r \rangle$ contains no nonzero polynomials of degree less than d .

Now, every element of $\langle g_1, \dots, g_r \rangle$ is of the form

$$h_1 g_1 + \dots + h_r g_r, \quad h_i \in K[x_0, \dots, x_n]$$

so the map F_k which takes a tuple $(h_1, \dots, h_r)$ to the corresponding element is surjective from $(K[x_0, \dots, x_n]_{k-d})^r \rightarrow K[x_0, \dots, x_n]_k$. This map is K -linear, so if it is surjective then its rank is $\dim K[x_0, \dots, x_n]_k$. This is the number of partitions of k into $n+1$ intervals (to see this, note that any homogeneous monomial $x_0^{k_0} \dots x_n^{k_n}$ satisfies $k_0 + \dots + k_n = k$ which is a partition). By stars and bars, it has dimension

$$\binom{n+k}{n} = \binom{n+k}{k}$$

By linear algebra, this implies that one of the minors of size $\binom{n+k}{k}$ is nonzero. The minors are homogeneous polynomial equations in the coefficients g_i , which are correspondingly homogeneous polynomials in the coordinates of a . So $a \notin \pi(Z)$ if and only if a lies outside the collective projective vanishing locus of polynomial conditions, and therefore $\pi(Z)$ is a closed set. $\square$

Corollary 3.26

$\pi : \mathbb{P}^n \times Y \rightarrow Y$ is closed for any variety Y .

Proof. First we take the case where Y is affine in $\mathbb{A}^m$. Then for any $Z \subseteq \mathbb{P}^n \times Y$ closed, Z can be embedded in $\mathbb{P}^n \times \mathbb{A}^m \subseteq \mathbb{P}^n \times \mathbb{P}^m$. Let $\bar{Z}$ be the closure of Z in $\mathbb{P}^n \times \mathbb{P}^m$. By the previous theorem, the projection $\pi' : \mathbb{P}^n \times \mathbb{P}^m \rightarrow \mathbb{P}^m$ is closed, so

$$\pi'(\bar{Z}) \subseteq \mathbb{P}^m$$

is closed. Then returning to $\mathbb{P}^n \times Y$, since Z is closed in $\mathbb{P}^n \times Y$ we have

$$\pi(Z) = \pi'(\bar{Z} \cap (\mathbb{P}^n \times Y)) = \pi'(\bar{Z}) \cap Y$$

which is closed in Y .

Now, in the general case, we can cover any variety Y by affine open sets Y_i . Then any closed set $Z \subseteq \mathbb{P}^n \times Y$ satisfies

$$\pi(Z) \cap Y_i = \pi(Z \cap (\mathbb{P}^n \times Y_i))$$

which is closed in Y_i by the affine case. So $\pi(Z)$ is closed when checking on an open cover, which means it is closed in Y . $\square$

We are now ready to state the definition of completeness.

Definition 3.19

A variety X is called a **complete variety** if the projection map $\pi : X \times Y \rightarrow Y$ is closed for any variety Y .

The previous corollary shows that $\mathbb{P}^n$ is complete, which generalizes the fact that $\mathbb{P}_{\mathbb{C}}^n$ is compact in the Euclidean topology.

Example 3.9

We show that $\mathbb{A}^1$ is not complete. Informally, this is because it is “missing” the points at infinity in its projective closure. For instance, consider the closed hyperbola $V(x_1x_2 - 1) \subseteq \mathbb{A}^1 \times \mathbb{A}^1$. Taking the projection, its image is $\mathbb{A}^1 \setminus \{0\}$, which is not closed. If we take the projective closure, then we have $V_p(x_1x_2 - x_0^2) \subseteq \mathbb{P}^1 \times \mathbb{A}^1$ which projects to $\mathbb{A}^1$.

Proposition 3.27

A closed subvariety of a complete subvariety is complete.

Proof. Let $X' \subseteq X$ be a closed subvariety of a complete subvariety. Then for Y a variety, we have the following diagram:

$$\begin{array}{ccc} X' \times Y & \hookrightarrow & X \times Y \\ \pi' \downarrow & & \downarrow \pi \\ Y & \xleftarrow{\cong} & Y \end{array}$$

π is a closed map since X is complete, and the inclusion $X' \times Y \hookrightarrow X \times Y$ is a closed embedding. Finally, the isomorphism is obviously closed, so π' is the composition of closed maps, hence closed $\square$

In particular, every projective variety is complete. We note that there exist complete varieties which are not projective, but they are very hard to construct and not encountered in practice.

Corollary 3.28

If $f : X \rightarrow Y$ is a morphism of varieties and X is complete then $f(X) \subseteq Y$ is a complete closed subvariety. Also, f is a closed map.

Proof. X is complete so the projection $\pi : X \times Y \rightarrow Y$ is a closed map. Since Y is a variety, the graph $\Gamma_f = \{(x, f(x)) : x \in X\} \subseteq X \times Y$ is closed. Therefore $f(X) = \pi(\Gamma_f)$ is closed in Y .

To see that $f(X)$ is complete, let Y' be any other variety, and let $Z \subseteq f(X) \times Y'$ be closed. We lift to $X \times Y'$ via the morphism $\psi = (f, \text{id}_{Y'})$. Thus we have the following diagram:

$$\begin{array}{ccc} \psi^{-1}(Z) & \subseteq & X \times Y' \\ \downarrow \psi & & \downarrow \psi \quad \curvearrowright \quad \pi \\ Z & \subseteq & f(X) \times Y' \\ \downarrow \pi' & & \downarrow \pi' \quad \curvearrowleft \\ \pi'(Z) & \subseteq & Y' \end{array}$$

Since ψ is surjective, we get

$$\pi'(Z) = \pi'(\psi(\psi^{-1}(Z))) = \pi(\psi^{-1}(Z))$$

$\psi^{-1}(Z)$ is closed since ψ is a morphism, hence continuous, and π is a closed map because X is complete. So π' is closed and thus $f(X)$ is complete.

To see that f is a closed map, we know that for any $V \subseteq X$ closed, V itself is complete. So $f(V)$ is the image of the composition $f \circ i$ with i the closed embedding morphism. This itself is a morphism so by the first part, $f(V)$ is closed. $\square$

Example 3.10: Exercise

Generalize the construction of the affine line with two origins by gluing two copies of $\mathbb{P}^1$ along $\mathbb{P}^1 \setminus \{0\}$ to get a projective line with two origins X . Show that the image of the embedding $\mathbb{P}^1 \rightarrow X$ is not closed in X .

Recall that in complex analysis, we know that if a holomorphic function from the Riemann sphere $\mathbb{P}^1_{\mathbb{C}} \rightarrow \mathbb{C}$ is bounded, then it is constant. We have a similar analogue in the algebraic projective setting.

Corollary 3.29

Let X be a connected complete variety. Then the set of regular functions is the constant functions $\mathcal{O}_X(X) = K$.

Proof. Let $\varphi \in \mathcal{O}_X(X)$. Then $\varphi : X \rightarrow \mathbb{A}^1$ is a morphism, and by embedding $\mathbb{A}^1 \subseteq \mathbb{P}^1$, we get a map $\varphi : X \rightarrow \mathbb{P}^1$ with $\infty \notin \varphi(X)$. By the above, $\varphi(X)$ is a closed subset of $\mathbb{P}^1$, so it is relatively closed in $\mathbb{A}^1$, meaning it is a finite set or all of $\mathbb{A}^1$. But in the second case, $\mathbb{A}^1$ is not closed in $\mathbb{P}^1$ so this cannot be the case. Since X is connected, the image is connected, and the only connected finite set is a single point. So φ is constant. $\square$

We saw that when $f \in K[x_0, \dots, x_n]$ is degree 1 and homogeneous, $X = V_p(f)$ is a linear hypersurface. In fact, we can show that this is isomorphic to the particular linear hypersurface $V_p(x_0)$. We show this using the Veronese embedding.

Definition 3.20

Let $n, d > 0$ be integers and let $f_0, \dots, f_N$ be the set of degree d monomials in $K[x_0, \dots, x_n]$ (where $N = \binom{n+d}{n} - 1$ by a stars and bars argument). Then the **Veronese embedding** of degree d is the map

$$F = F_{n,d} : \mathbb{P}^n \rightarrow \mathbb{P}^N, x \mapsto (f_0(x) : \dots : f_N(x))$$

Note that the f_j do not all vanish simultaneously since the monomials $x_0^d, \dots, x_n^d$ do not all vanish simultaneously. This is a map with polynomial coordinates that do not simultaneously vanish, so this is a morphism. Since $\mathbb{P}^n$ is a complete variety and $\mathbb{P}^N$ a variety, the image $X = F(\mathbb{P}^n) \subseteq \mathbb{P}^N$ is a complete closed subvariety, and F is a closed map.

Proposition 3.30

The Veronese embedding is an isomorphism onto its image (and therefore a closed embedding).

Proof. We can check the inverse morphism on an open cover. Let U_i be the open set $\{x_i \neq 0\}$, hence $x_i^d \neq 0$. Setting $x_i = 1$, we get affine coordinates $x_0, \dots, x_{i-1}, x_{i+1}, \dots, x_n$ on U_i . On U_i , the inverse morphism has coordinates (for $j \neq i$)

$$x_j = \frac{x_j x_i^{d-1}}{x_i^d}$$

which is the quotient of coordinate functions where the denominator does not vanish, so F is an isomorphism. $\square$

This embeds $\mathbb{P}^n$ into a bigger space where the degree d conditions are translated into linear conditions.

Example 3.11

For $n = 1$, we have

$$F_{1,d} : \mathbb{P}^1 \rightarrow \mathbb{P}^d, (x_0 : x_1) \mapsto (x_0^d : x_0^{d-1}x_1 : \dots : x_0x_1^{d-1} : x_1^d)$$

which is called the rational normal curve.

Proposition 3.31

The image of the Veronese embedding is the vanishing locus of the equations

$$X_I \cdot X_J = X_K \cdot X_L$$

where I, J, K, L are multiindices, X_α is the monomial $x_0^{\alpha_0} \dots x_n^{\alpha_n}$, and $I + J = K + L$ as multiindices.

The above is a quadratic condition, so we see that any projective variety is isomorphic to one (possibly in higher dimension) cut out by quadratic conditions.

Corollary 3.32

Let $X \subseteq \mathbb{P}^n$ be a projective variety and $f \in S(X)$ be homogeneous and nonconstant. Then $X \setminus V(f)$ is an affine variety.

Proof. First, in the case $X = \mathbb{P}^n$ and f is a linear polynomial $f(x) = a_0x_0 + \dots + a_nx_n$, we

can extend $(a_0, \dots, a_n)$ to a basis of K^{n+1} , and write

$$A = \begin{bmatrix} a_0 & \cdots & a_n \\ * & \cdots & * \\ \vdots & \ddots & \vdots \\ * & \cdots & * \end{bmatrix}$$

so that

$$V_p(f) = \{x : (Ax)_0 = 0\} \cong \{x : x_0 = 0\}$$

where the isomorphism follows since A is a projective automorphism. So $\mathbb{P}^n \setminus V_p(f) \cong \mathbb{P}^n \setminus V_p(x_0) \cong \mathbb{A}^1$ is affine. In the general case, where f is of degree d , the Veronese embedding $F_{n,d}$ takes f to a linear polynomial $g = F_{n,d} \circ f$. Then $X \setminus V(f)$ is a closed subset of $\mathbb{P}^n \setminus V(f)$, which is isomorphic to $Y \setminus V(g)$, which is a closed subset of $\mathbb{P}^N \setminus V(g)$, which we just showed is affine. So we are done. $\square$

3.4 Grassmannians

We now offer an introduction to a generalization of projective space which will later enable us to perform blowup modifications on projective varieties – essentially, to “zoom in,” for instance by looking at the tangent space.

Definition 3.21

For $n > 0$ and $0 \leq k \leq n$, the **Grassmannian** $\text{Gr}(k, n)$ is the set of k -dimensional linear subspaces of K^{n+1} .

By definition, this generalizes projective space as $\text{Gr}(1, n) = \mathbb{P}^{n-1}$. We will show that the Grassmannian can be embedded as a projective variety inside $\mathbb{P}^N$ for some large value of N . To do this we will need to be able to write down coordinates which are unique up to scaling. If we take any $L \in \text{Gr}(k, n)$ and pick a basis $v_1, \dots, v_k$ of L , then the matrix

$$\begin{bmatrix} v_{1,1} & \cdots & v_{k,1} \\ \vdots & \ddots & \vdots \\ v_{1,n} & \cdots & v_{k,n} \end{bmatrix}$$

is unique up to column operations, which correspond to right multiplication by elements of $\text{GL}_k(K)$. For any maximal $k \times k$ minor of this matrix, right multiplication by an element $S \in \text{GL}_k(K)$ just serves to scale the minor by $\det(S) \neq 0$. Thus the maximal minors of this matrix are unique up to homogeneous scaling. Moreover they are not all zero since this matrix has rank k .

Definition 3.22

For a linear subspace $L \in \text{Gr}(k, n)$, the vector p_L of all $k \times k$ maximal minors are called the **Plücker coordinates** of L .

We can more generally write this without respect to a basis using the language of tensor products and alternating maps.

Definition 3.23

A multilinear map $f : V^k \rightarrow W$ for two K -vector spaces V, W and $k \in \mathbb{N}$ is called an **alternating map** if $f(v_1, \dots, v_k) = 0$ whenever $v_i = v_j$ for distinct i, j .

We remark that by linearity, and expanding out the quantity

$$f(v_1, \dots, v_i + v_j, \dots, v_i + v_j, \dots, v_k)$$

we recover the more familiar property that switching two arguments reverses the sign of f . More generally,

Proposition 3.33

If $f : V^k \rightarrow W$ is alternating and $\sigma \in S_k$ is a permutation, then

$$f(v_{\sigma(1)}, \dots, v_{\sigma(k)}) = \text{sign}(\sigma) f(v_1, \dots, v_k)$$

Definition 3.24

A k -fold **alternating tensor product** of a K -vector space V is a K -vector space T together with a k -fold alternating map $\tau : V^k \rightarrow T$ such that for any k -fold alternating map $f : V^k \rightarrow W$ there is a unique linear map $g : T \rightarrow W$ such that $f = g \circ \tau$:

$$\begin{array}{ccc} & & T \\ & \nearrow \tau & \downarrow \exists! g \\ V^k & \xrightarrow{f} & W \end{array}$$

As usual, alternating tensor products are unique up to unique isomorphism. Thus such a product is also denoted $T = \bigwedge^k V$ and $\tau(v_1, \dots, v_k) = v_1 \wedge \dots \wedge v_k$.

Proposition 3.34

For any V, k , an alternating tensor product exists.

Proof. We take T to be the k -fold tensor product $V \otimes \dots \otimes V$, quotiented by the linear subspace L which is generated by elements of the form $v_1 \wedge \dots \wedge v_k$ where $v_i = v_j$ for some $i \neq j$. $\square$

As with alternating maps, we have that

$$v_{\sigma(1)} \wedge \dots \wedge v_{\sigma(k)} = \text{sign}(\sigma) v_1 \wedge \dots \wedge v_k$$

Moreover, a basis for $\bigwedge^k V$ is given by vectors of the form

$$e_{i_1} \wedge \cdots \wedge e_{i_k}$$

where each e_{i_j} is chosen from a basis $\{e_1, \dots, e_n\}$ of V , and each is distinct. Since permutation only performs scalar multiplication, we can also assume that $i_1 < \dots < i_k$, which then gives that

$$\dim \bigwedge^k V = \binom{n}{k}$$

Finally, we note that $\bigwedge^0 V \cong K$, $\bigwedge^1 V \cong V$, and by the dimension formula, $\dim \bigwedge^n K^n = 1$, so it is isomorphic to K . In particular, the determinant gives a canonical isomorphism.

Now, select vectors $v_1, \dots, v_k \in K^n$ for $0 \leq k \leq n$, and suppose they have coordinates $v_i = \sum_j a_{ij} e_j$ in some basis $\{e_j\}$. Let us determine how to write the coordinates of $v_1 \wedge \cdots \wedge v_k$ in terms of the $e_{i_1} \wedge \cdots \wedge e_{i_k}$. By multilinearity and regrouping so every term has increasing indices, we have

$$\begin{aligned} v_1 \wedge \cdots \wedge v_k &= \sum_{i_1, \dots, i_k} a_{1i_1} \cdots a_{ki_k} e_{i_1} \wedge \cdots \wedge e_{i_k} \\ &= \sum_{i_1 < \cdots < i_k} \left(\sum_{\sigma \in S_k} \text{sign}(\sigma) a_{1i_{\sigma(1)}} \cdots a_{ki_{\sigma(k)}} \right) e_{i_1} \wedge \cdots \wedge e_{i_k} \end{aligned}$$

This coefficient is precisely the Leibniz determinant formula for the $k \times k$ submatrix with rows $i_1, \dots, i_k$ selected from the following matrix:

$$\begin{bmatrix} a_{11} & \cdots & a_{k1} \\ \vdots & \ddots & \vdots \\ a_{1n} & \cdots & a_{kn} \end{bmatrix}$$

Thus we see that the coefficients of $v_1 \wedge \cdots \wedge v_k$ are precisely the maximal minors of the above matrix, which are the same as the Plücker coordinates we found earlier.

Lemma 3.35

$v_1 \wedge \cdots \wedge v_k = 0 \in \bigwedge^k V$ if and only if $v_1, \dots, v_k$ are linearly dependent in V .

Proof. $v_1, \dots, v_k$ are linearly dependent if and only if all $k \times k$ maximal minors are zero, which is equivalent to all coordinates of $v_1 \wedge \cdots \wedge v_k$ being zero, hence equivalent to $v_1 \wedge \cdots \wedge v_k$ being zero. $\square$

Lemma 3.36

If $v_1, \dots, v_k$ and $w_1, \dots, w_k$ are two sets of linearly independent vectors in K^n , then $v_1 \wedge \cdots \wedge v_k, w_1 \wedge \cdots \wedge w_k$ are linearly dependent in $\bigwedge^k V$ if and only if they span

the same k dimensional linear subspaces of V .

Proof. We remark that by the previous lemma, since both sets are linearly independent, they define k dimensional subspaces and their wedge products are nonzero.

($\Rightarrow$) If

$$v_1 \wedge \cdots \wedge v_k = \lambda w_1 \wedge \cdots \wedge w_k$$

for some $\lambda \in K$, then taking the wedge product against w_i , we have

$$w_i \wedge v_1 \wedge \cdots \wedge v_k = 0$$

so $\{v_1, \dots, v_k, w_i\}$ is linearly dependent. Since $\{v_1, \dots, v_k\}$ is linearly independent, w_i lies in its span, which is true for all i , so

$$\text{span}(w_1, \dots, w_k) \subseteq \text{span}(v_1, \dots, v_k)$$

and the reverse is true as well.

($\Leftarrow$) If

$$\text{span}(w_1, \dots, w_k) \subseteq \text{span}(v_1, \dots, v_k)$$

then there is a matrix $B \in \text{GL}_k(K)$ which takes

$$w_i = \sum_j b_{ij} v_j$$

from which it follows that

$$w_1 \wedge \cdots \wedge w_k = \sum_{j_1, \dots, j_k} b_{1j_1} \cdots b_{kj_k} v_1 \wedge \cdots \wedge v_k = \det B v_1 \wedge \cdots \wedge v_k$$

by the same regrouping argument as above. $\det B \neq 0$, so we are done. $\square$

Definition 3.25

Let $0 \leq k \leq n$. The **Plücker embedding** is the map

$$f : \text{Gr}(k, n) \rightarrow \mathbb{P}^{\binom{n}{k}-1}$$

which takes $L \in \text{Gr}(k, n)$ spanned by $v_1, \dots, v_k$ to the coordinates of $v_1 \wedge \cdots \wedge v_k$ with respect to some basis $\{e_j\}$.

By the lemma above, these are independent of the choices of the v_i and the e_j up to homogeneous scaling. Also, since $v_1, \dots, v_k$ are linearly independent, they are not all zero. Finally, the lemma above also tells us that this map is injective.

Example 3.12

For $\text{Gr}(1, n)$, we have

$$f(\text{span}(a_1 e_1 + \dots + a_n e_n)) = (a_1 : \dots : a_n) \in \mathbb{P}^{n-1}$$

so $\text{Gr}(1, n) = \mathbb{P}^{n-1}$ as we expected.

Thus we have seen that we have a somewhat natural embedding of the Grassmannian into projective space, but this is only useful if we can show that it is open or closed. We will show that it is a closed subset of $\mathbb{P}^N$, which means the Plücker embedding defines the structure of a projective variety on $\text{Gr}(k, n)$. Here, we note that the image of a linear subspace under the Plücker embedding is given by the pure tensor

$$v_1 \wedge \dots \wedge v_k$$

Conversely, we know that every such nonzero pure tensor has $v_1, \dots, v_k$ linearly independent, so their span is the preimage of this pure tensor. Therefore the Grassmannian, embedded in $\mathbb{P}^N$, is the set of pure tensors in $\bigwedge^k V$ (while $\bigwedge^k V$ in general consists of linear combinations thereof).

Lemma 3.37

Let $\omega \in \bigwedge^k K^n$ be nonzero with $k < n$, and consider the map

$$f : K^n \rightarrow \bigwedge^{k+1} K^n, v \mapsto v \wedge \omega$$

Then f has rank at least $n - k$, with equality if and only if ω is a pure tensor.

Proof. Pick a basis $v_1, \dots, v_r$ for $\ker f$ (where for dimension reasons we have $r = n - \text{rank } f$). Extend this to a basis $v_1, \dots, v_n$ of K^n . Then write

$$\omega = \sum_{i_1 < \dots < i_k} a_{i_1, \dots, i_k} v_{i_1} \wedge \dots \wedge v_{i_k}$$

Since the v_i , $1 \leq i \leq r$, are in the kernel of f , we have that

$$0 = v_i \wedge \omega = \sum_{i_1 < \dots < i_k} a_{i_1, \dots, i_k} v_i \wedge v_{i_1} \wedge \dots \wedge v_{i_k}$$

For each term, if $i \in \{i_1, \dots, i_k\}$ then the corresponding tensor for that term is zero. If $i \notin \{i_1, \dots, i_k\}$ then the tensor is nonzero, so since the LHS is zero, by linear independence we have that the coefficients $a_{i_1, \dots, i_k}$ are zero $i \notin \{i_1, \dots, i_k\}$. This is true for all $i \leq r$ so the only nonzero coefficients are those such that $\{1, \dots, r\} \subseteq \{i_1, \dots, i_k\}$. Since ω is nonzero there must be at least one nonzero coefficient, so we have $k \geq r$, which implies that $\text{rank}(f) \geq n - k$. Moreover, if there is equality then only the coefficient $a_{1, \dots, k}$ can be nonzero so this is a pure tensor.

Conversely, if ω is a pure tensor $\omega_1 \wedge \cdots \wedge \omega_k$ then each ω_j is in the kernel of f (and they are linearly independent), so we have

$$\dim \ker(f) \geq k \implies \text{rank}(f) \leq n - k$$

so we have both inequalities and thus $\text{rank}(f) = n - k$. $\square$

Corollary 3.38

Under the Plücker embedding, $\text{Gr}(k, n)$ is a closed subset of $\mathbb{P}^{\binom{n}{k}-1}$.

Proof. The statement is true for $n = k$ as the Grassmannian is a single point. So assume $n < k$, as we need this for the previous result. As we discussed above, our previous results have shown that the image of the Plücker embedding is the set of all (nonzero) pure tensors in $\bigwedge^k K^n$ (up to equivalence classes by homogeneous scaling).

Then, the previous lemma says that this condition is equivalent to the map $f : v \mapsto v \wedge \omega$ having rank $n - k$. Since the rank is always at least $n - k$, it is enough to have the condition $\text{rank}(f) \leq n - k$. This is the condition that all $n - k + 1$ minors of the matrix of f vanish. This is a polynomial in the entries of the matrix of f , which in turn can be written as polynomials in the Plücker coordinates of ω . Therefore $\text{Gr}(k, n)$ can be defined by polynomial vanishing conditions, and is a closed subset. $\square$

The Plücker embedding indirectly gives an affine cover of $\text{Gr}(k, n)$, by considering the affine covers of $\mathbb{P}^{\binom{n}{k}-1}$. We can explicitly understand what these patches look like.

Proposition 3.39

$\text{Gr}(k, n)$ admits an open cover by spaces isomorphic to $\mathbb{A}^{(n-k) \times k}$.

Proof. Without loss of generality, we will consider the patch

$$U_0 = \left(\mathbb{P}^{\binom{n}{k}-1} \setminus V_p(x_0) \right) \cap \text{Gr}(k, n)$$

This is the subset of $\text{Gr}(k, n)$ where the coordinate of

$$e_1 \wedge \cdots \wedge e_k$$

is nonzero. In other words, based on our descriptions of the elements of $\text{Gr}(k, n)$ as the column spans of matrices with the correct maximal minors, we know that any element of U_0 is represented as the column span of a block matrix of the form

$$\begin{bmatrix} A \\ B \end{bmatrix}$$

where A is $k \times k$ and B is $(n-k) \times k$. The coordinate of $e_1 \wedge \cdots \wedge e_k$ is the determinant of A , so A is invertible. In particular, since the column span is invariant under right multiplication by elements of $\mathrm{GL}_k(K)$, we know that this is the same element as the one represented by

$$\begin{bmatrix} I_k \\ BA^{-1} \end{bmatrix}$$

Since the I_k block gives a nonzero coordinate, the matrix B may be chosen freely. Thus, U_0 is parameterized by

$$U_0 = \left\{ \text{ColumnSpan} \begin{bmatrix} I_k \\ C \end{bmatrix} : C \in \mathbb{A}^{(n-k) \times k} \right\}$$

Moreover, the maximal minors of

$$\begin{bmatrix} I_k \\ C \end{bmatrix}$$

are polynomials in the entries of C , so the map

$$f : \mathbb{A}^{(n-k) \times k} \rightarrow U_0, C \mapsto \begin{bmatrix} I_k \\ C \end{bmatrix}$$

is a morphism. Moreover, the minor of

$$\begin{bmatrix} I_k \\ C \end{bmatrix}$$

given by choosing all the rows of I_k except the i th, and the j th row of C , is exactly C_{ij} (up to sign, and we can permute the rows based on the parities of i, j to make sure this works out). Thus the map is also injective, and the inverse map is a projection since the C_{ij} are contained as Plücker coordinates of the block matrix. Therefore U_0 is isomorphic to $\mathbb{A}^{(n-k) \times k}$, and $\mathrm{Gr}(k, n)$ is covered by copies of $\mathbb{A}^{(n-k) \times k}$. $\square$

Corollary 3.40

$\mathrm{Gr}(k, n)$ is irreducible and of dimension $(n-k)k$.

Proof. Any two of the patches has nonempty intersection, and in particular there is a matrix which lies in all the patches. Each patch is irreducible of dimension $(n-k)k$, so for topological reasons it follows that $\mathrm{Gr}(k, n)$ is of dimension $(n-k)k$ and is irreducible as well. $\square$

Proposition 3.41

For $0 \leq k \leq n$, $\mathrm{Gr}(k, n) \cong \mathrm{Gr}(n-k, n)$.

Proof. The set theoretic map $L \mapsto L^\perp$ is a bijection between $\mathrm{Gr}(k, n)$ and $\mathrm{Gr}(n-k, n)$. Also, the inverse map is also $\tilde{L} \mapsto \tilde{L}^\perp$, so it suffices to show that the forward map is a morphism. We can show this on an affine patch, so consider the patch where L is parameterized by

$$\begin{bmatrix} I_k \\ C \end{bmatrix}$$

Note that we have

$$\begin{bmatrix} -C & I_{n-k} \end{bmatrix} \begin{bmatrix} I_k \\ C \end{bmatrix} = 0$$

as a matrix, so the row span of $\begin{bmatrix} -C \\ I_{n-k} \end{bmatrix}$ is orthogonal to L . Since the row span is linearly independent (as there is a nonzero coordinate given by the I_{n-k} minor), we get that the row span lies in $L^\perp$, and for dimension reasons $L^\perp$ is precisely this row span. Moreover, the Plücker coordinates of $L^\perp$ are the maximal minors of $\begin{bmatrix} -C & I_{n-k} \end{bmatrix}$ which are polynomials in the entries of C , so this is a morphism and the desired isomorphism is found. $\square$

Chapter 4

Birational Geometry

4.1 Birational Maps and Blowing Up

In birational geometry we wish to study the ways that varieties are related to each other, in particular those which look similar on a dense open set. For instance, we know that $\mathbb{P}^n \times \mathbb{P}^m$ is not isomorphic to $\mathbb{P}^{n+m}$, but they both contain a dense open subset $\mathbb{A}^n \times \mathbb{A}^m \cong \mathbb{A}^{n+m}$. Or, we know that $V(y^2 - x^3)$ is not isomorphic to $\mathbb{A}^1$ itself, but removing the origin we do have

$$V(y^2 - x^3) \setminus \{(0, 0)\} \cong \mathbb{A}^1 \setminus \{0\}$$

via the map $t \mapsto (t^2, t^3)$, $(x, y) \mapsto y/x$. We will first describe this kind of classification and then make a construction that allows us to study these types of relationships.

Definition 4.1

A **rational map** between irreducible varieties X, Y is a morphism $f : U \rightarrow Y$ where U is a nonempty open subset of X . Such a rational map is denoted $f : X \dashrightarrow Y$. Two rational maps $f_1, f_2 : X \dashrightarrow Y$ defined on U_1, U_2 are equivalent if there is U nonempty contained in $U_1 \cap U_2$ where f_1, f_2 agree as morphisms.

We note that if two such maps are equivalent, then since U is dense in $U_1 \cap U_2$, and the locus where $f_1 = f_2$ agree is closed in $U_1 \cap U_2$, it always suffices to check that $f_1|_{U_1 \cap U_2} = f_2|_{U_1 \cap U_2}$. As defined, two rational maps need not have a well defined composition. For instance, if $f : \mathbb{A}^1 \dashrightarrow \mathbb{A}^1, x \mapsto 0$ and $g : \mathbb{A}^1 \dashrightarrow \mathbb{A}^1, x \mapsto 1/x$, then g can never be defined on the image of f , so we cannot compose $g \circ f$.

Definition 4.2

A rational map $f : X \dashrightarrow Y$ with X, Y irreducible is **dominant** if $\text{im } f$ is dense in Y .

Note that if $f : X \dashrightarrow Y$ is dominant and $g : Y \dashrightarrow Z$ is a rational map defined on $V \subseteq Y$, then $g \circ f : X \dashrightarrow Z$ is defined on $f^{-1}(V)$, which is nonempty since f is dominant.

Definition 4.3

A rational map $f : X \dashrightarrow Y$ with X, Y irreducible is a **birational map** if f is dominant and there is $g : Y \dashrightarrow X$ dominant with $g \circ f = \text{id}_X, f \circ g = \text{id}_Y$ under equivalence of rational maps. In this case X, Y are said to be birational.

If we restrict our attention to dominant maps, then composition is always well defined and the composition is also dominant. Thus, the category of irreducible varieties with morphisms given by dominant rational maps forms a category $\text{Var}_K^{\text{dom}}$. Birationality gives us a simple condition to check that two varieties have isomorphic subsets.

Lemma 4.1

Two irreducible varieties X, Y are birational if and only if they contain isomorphic nonempty open subsets.

Proof. ($\Leftarrow$) If $U \subseteq X, V \subseteq Y$ are isomorphic via $f : U \rightarrow V, f^{-1} : V \rightarrow U$, then f, f^{-1} are both rational maps by definition. They are also dominant as the images U, V are dense by assumption. Hence X, Y are birational.

($\Rightarrow$) If $f : X \dashrightarrow Y$ is defined on $U \subseteq X$ and $g = f^{-1} : Y \dashrightarrow X$ is defined on $V \subseteq Y$, then both compositions are defined on

$$f^{-1}(g^{-1}(U)), g^{-1}(f^{-1}(V))$$

respectively, so these two sets are isomorphic (and nonempty open, hence dense). $\square$

Appendix A

Results in Commutative Algebra

This appendix is for results in commutative algebra which are not explicitly covered in the lecture notes. As in the main notes, every ring is assumed to be commutative with unity.

A.1 Hilbert's Basis Theorem

This result is taken from Eisenbud's *Commutative Algebra: with a View toward Algebraic Geometry*. One consequence of the basis theorem is that every ideal in a polynomial ring is finitely generated. This is an important basic result which is used in the proof that every variety is the finite union of irreducibles.

Definition A.1

A ring is said to be **Noetherian** if every strictly ascending chain of ideals terminates, or equivalently if every ideal is finitely generated.

Proposition A.1

These two definitions are equivalent.

Proof. For an ideal $I \subseteq R$, take $r_1 \in R$, $r_2 \in R \setminus \langle r_1 \rangle$, $r_3 \in R \setminus \langle r_1, r_2 \rangle$, and so on. Then $\langle r_1 \rangle \subseteq \langle r_1, r_2 \rangle \subseteq \dots$ is strictly ascending, so it terminates. It only terminates if $\langle r_1, \dots, r_k \rangle = I$, so I is finitely generated.

Take $I_1 \subseteq I_2 \subseteq \dots$. Then $\bigcup I_i$ is an ideal, and it is finitely generated. These generators are all included in some I_k , so the chain terminates. $\square$

Example A.1

Fields only have themselves and the zero ideal as ideals, so they are Noetherian. By the Chinese Remainder Theorem, $\mathbb{Z}$ is Noetherian, since any ideal is generated by the gcd.

Example A.2

Every ring is finitely generated as an ideal (since $R = \langle 1 \rangle$). However, we can also consider rings which are finitely generated as $\mathbb{Z}$ -modules. This means that every element is a finite sum over the generating set S of the form

$$n_1 s_1 + \dots + n_k s_k, \quad n_i \in \mathbb{Z}, s_i \in S$$

As a result of the basis theorem, we will see that any ring which is finitely generated as a $\mathbb{Z}$ -module, then the ring itself is also Noetherian.

Theorem A.2: Hilbert's Basis Theorem

If R is Noetherian then $R[x]$ is as well.

The basic point of the proof is that the leading terms of polynomials effectively act like the ring itself, when they share the same degree. For a polynomial

$$f = a_n x^n + \dots + a_1 x + a_0$$

a_n is called the leading coefficient and $a_n x^n$ the leading term.

Proof. Pick an ideal I of $R[x]$. Let f_1 be any nonzero polynomial of lowest degree in I . Recursively choose f_i to be a nonzero polynomial of lowest degree in $I \setminus (f_1, \dots, f_{i-1})$, terminating if this is ever empty. If it does not terminate, then we have infinitely many leading coefficients: denote the leading coefficient of f_j by a_j . Then $(a_1, a_2, \dots)$ is an ideal in R , and is in particular finitely generated by $a_1, \dots, a_m$ for some m . Then we can write

$$a_{m+1} = \sum_{j=1}^m a_j u_j$$

for some $u_j \in R$. We can then contradict our lowest degree construction by showing that the leading term of f_{m+1} is actually in $(f_1, \dots, f_m)$. By increasing the degree of our polynomials, we can cancel the leading term: define

$$g = \sum_{j=1}^m u_j a f_j x^{\deg f_{m+1} - \deg f_j}$$

which has leading term equal to $a_{m+1} x^{\deg f_{m+1}}$. Therefore $f_{m+1} - g$ has degree strictly less than f_{m+1} , but is also not $(f_1, \dots, f_m)$, contradicting our choice of f_{m+1} . $\square$

A.2 Primary Decompositions of Noetherian Rings

This is Atiyah-MacDonald Chapter 7.

Lemma A.3

If R is Noetherian then every ideal is a finite intersection of irreducible ideals.

Proof. If there are some ideals that are not a finite intersection, then the set of such ideals is nonempty and has a maximal element $\mathfrak{a}$. This is reducible so it is the intersection of $\mathfrak{a} = \mathfrak{b} \cap \mathfrak{c}$ which both strictly contain $\mathfrak{a}$. But then $\mathfrak{b}, \mathfrak{c}$ are finite intersections of irreducible ideals and so is $\mathfrak{a}$. $\square$

Lemma A.4

In a Noetherian ring every irreducible ideal is primary.

Proof. We can quotient out by the ring, so this is equivalent to the statement that in any Noetherian ring, the zero ideal is primary if it is irreducible. Suppose the zero ideal is irreducible. Then take x, y with $y \neq 0$ and $xy = 0$. Then we need to show that $x^n = 0$ for some n .

Consider the chain of annihilators

$$\text{Ann}(x) \subseteq \text{Ann}(x^2) \subseteq \dots$$

The ring is Noetherian so there is n such that $\text{Ann}(x^n) = \text{Ann}(x^{n+1})$. We claim that $(x^n) \cap (y) = 0$. Indeed, if $a \in (x^n) \cap (y)$ then $a \in (y)$ so $ax = 0$. If also $a \in (x^n)$ then $a = bx^n$, and $ax = bx^{n+1} = 0$ so $b \in \text{Ann}(x^{n+1}) = \text{Ann}(x^n)$ so $0 = bx^n = a$. Then $(y) \neq 0$ so $(x^n) = 0$, since 0 is irreducible. So 0 is primary. $\square$

The following follows immediately:

Theorem A.5

In a Noetherian ring, every ideal has a primary decomposition.

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